

Contest Segmentation, Recognition, and the Youth Talent Pipeline

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Abstract

Schools and youth programs sort participants into ability groups and recognize top performers within each. Tracking bundles peers, instruction, and local rank; this paper isolates the rank-and-recognition channel on the margin compulsory schooling hides—whether participants come back at all. I use administrative panels from Utah high-school mountain-bike racing (2018–2025) and five other state leagues, where ability-group assignment leaves riders of similar speed facing different odds of standing on a podium. Instrumenting competitive success with the recognition a rider’s assigned field makes attainable—using only strictly predetermined ability and shock information—each additional top-five finish raises a returning rider’s next-season participation by about two percentage points, a magnitude on which three independent designs agree; first-year riders show no detectable response, and outside Utah the estimates are positive but underpowered. What differs across leagues is the supply of recognition that structure creates: subdividing fields roughly doubles the top-five finishes awarded—a recognition-supply policy worth up to 1.2 points of annual persistence. Because early performance only weakly predicts eventual success, retaining marginal participants has option value.

Keywords: Tracking, ability grouping, contest design, ordinal rank, recognition, persistence, youth sports, instrumental variables, microsimulation

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1 Introduction

How should an organizer partition heterogeneous participants into ability groups, and how deeply should recognition reach within each? Every tracked classroom, gifted program, leveled course sequence, and youth league answers these questions, almost always by convention rather than by evidence. Two literatures bear directly on the answer and stop just short of it. The economics of tracking asks whether ability grouping raises *achievement* (Betts, 2011; Duflo, Dupas and Kremer, 2011; Card and Giuliano, 2016; Pop-Eleches and Urquiola, 2013; Abdulkadiroğlu, Angrist and Pathak, 2014), but the treatment it studies is a bundle: assignment changes a participant’s peers, the instruction aimed at them, and their *local rank*—the position they hold among the people they are compared with every day. The rank-effects literature has since shown that local rank, holding ability fixed, changes investment and later outcomes (Murphy and Weinhardt, 2020; Elsner and Isphording, 2017; Elsner, Isphording and Zölitz, 2021; Denning, Murphy and Weinhardt, 2023)—but it takes the partition as given. Neither literature treats the partition itself as the choice variable, and both study settings that hide the margin that matters most where participation is elective: not what a participant achieves, but whether they come back at all. A fifth grader cannot drop out of fifth grade; a student in an elective math circle, a music program, a coding club, or a sports league can simply not re-enroll, and in those settings continuation—not contemporaneous performance—is the outcome that determines who is still in the pipeline years later.

This paper isolates the rank-and-recognition channel of tracking in a setting built for the purpose. In interscholastic mountain-bike racing, every participant rides the same course on the same day, so the instruction-targeting channel of tracking is switched off by construction; peer quality is perfectly measured (in seconds) and can be controlled or instrumented; and the league re-enrolls riders every fall, so the participation margin is a real, repeated, observable choice for every participant. The league segments riders within grade and gender into ability divisions and publicly recognizes the top five finishers in each division at each race. Because assignment to divisions is governed by a written placement table that is far from

a deterministic ability sort—roughly half of assignment is unexplained by lagged ability—riders of nearly identical speed routinely face different local competition and correspondingly different odds of standing on a podium. The question is whether that manufactured local success causes riders to stay, and, if it does, what the league (or, by careful analogy, a school) should do with the levers it controls: the number and size of ability groups and the depth of recognition.

This setting is the cleanest available laboratory for a question education data structurally cannot answer, and it offers four things at once. First, the participation margin is a genuine annual choice observed for everyone, not a censored or compulsory one. Second, performance is measured in seconds rather than test scores, so ability and peer quality are observed essentially without error. Third, the assignment rule is written down, replicable, and leaves documented slack, which is the raw material for identification. Fourth, the design levers studied here—how many groups, how big, how deep the recognition—are levers the organizer actually adjusts season to season, so the counterfactuals are policies rather than thought experiments. The findings speak directly to the large class of elective educational activities in which dropout, not achievement, is the first-order margin (Stevenson, 2010; Barron, Ewing and Waddell, 2000).

I proceed in two linked steps. The first is causal: does locally generated competitive success affect whether athletes continue? I use three complementary designs, all leveraging the same primitive—a rider’s expected success depends on the field they are assigned to, which is only loosely tied to their own ability—and I present them in order of the cleanliness of their exclusion restrictions rather than their precision. The first instruments recognition with the strength of *external-school* competitors in a rider’s division, variation driven by other coaches’ choices; a fixed-reference variant evaluates each rider’s external field at a leave-one-out cell-median ability, so the instrument contains no function of the rider’s own estimated speed. The second simulates each rider-season’s expected recognition from pre-season speed and the actual composition of the field, using all the assignment slack at once;

for returning riders every input—the ability estimate and the race-shock distribution—comes from seasons strictly before the one being instrumented, demeaned within that window. That construction matters: residual pools that carry full-career location information inflate the estimate roughly threefold, and Section 7 reports the full correction ladder rather than hiding it. The third design exploits the sawtooth in predicted local position created by the placement table’s division-size cutoffs, in the spirit of Angrist and Lavy (1999); because the written rule reproduces only 57 percent of actual assignments, I present this cascade as an honest bounding design that rules out large effects but cannot on its own confirm the others—a failure that is itself informative, since it documents the placement discretion the first two designs exploit. For returning riders the per-finish effect on next-season participation lies between roughly +1 and +2.3 percentage points across all three designs (Section 7), and it survives conditioning directly on the rider’s full lagged history. The effect is a returning-rider phenomenon: a separate design for first-year riders, built only on their opening race, finds no detectable effect (−1.0 percentage points, standard error 1.0), and I keep the two populations separate rather than averaging them.

The second step is design evaluation, and it requires taking a stand—or openly declining to—on *what* carries the effect: the discrete recognition events (medals, podiums) or the continuous experience of local standing (where you finish relative to the field, race after race). The two imply different policy conclusions: recognition counts can be manufactured by the designer (more groups and deeper podiums mean more medals), whereas local standing is nearly zero-sum (someone must be beaten for someone to beat). I run the tests that could separate the mechanisms and report them all (Section 7). They do not settle it. A horse race that instruments both treatments at once favors continuous standing on the full sample, while the strictly predetermined returner design prices counts positively and standing negatively—the decomposition is not robust across instrument constructions. A pooled regression discontinuity at the fifth/sixth-place cutoff finds a precise zero at the single-race medal moment, so whatever matters is cumulative over the season rather than a medal

ceremony per se. And the per-finish effect declines as the share of the field recognized rises—a dilution pattern with a direct grade-inflation analogue (Butcher, McEwan and Weerapana, 2014). No simple version of either mechanism fits every one of these facts; I therefore remain agnostic about the mechanism and evaluate every counterfactual under both a recognition-count mapping and a standing-based mapping, reporting both side by side. The paper’s design conclusions are stated only where the mappings agree or with the divergence displayed.

The design question has force because of an option-value fact: early performance only weakly predicts who eventually populates the top track. About one-third of the riders who race varsity as seniors came from outside the top three deciles of early performance, and one in seven came from the bottom half. A long-run reduced form using only a rider’s *first race*—before any season-level feedback could operate—shows that early “rank luck” raises eventual varsity participation ($t = 7.2$), and a tier-split version shows the effect is present for non-elite entrants on the varsity margin but absent on the top-decile-ability margin: early success retains and advances ordinary riders into the top participation tier; it does not manufacture elite ability. Retention, in other words, has option value because the league cannot tell at entry who will be varsity-caliber at graduation.

The counterfactuals then quantify the design levers, priced at the corrected estimates and computed from cross-fitted baseline return probabilities on the non-senior population. Holding each season’s riders fixed, recognition-broadening policies (smaller fields, deeper podiums) raise the predicted return rate by up to 0.8 percentage points under the full-sample count mapping and up to 0.5 under the strict returner pricing, while a pure standing mapping implies near-zero aggregate effects of any lever—precisely because standing is zero-sum, so design can redistribute it but not create it. A dynamic microsimulation that follows real entering cohorts forward finds only modest compounding at these magnitudes: top-ten recognition—itself an extrapolation beyond the observed top-five support—raises the share of entrants who reach their senior year by about one percentage point, within a range from roughly zero (pricing the paper’s own dilution estimate into the simulation’s denser podiums

turns the gain slightly negative) to about two (zeroing the first-year response, consistent with the entrant design’s null, while returning riders respond at the strict estimate). The dynamic exercise now bounds rather than advertises: design levers move persistence, but at the corrected price the compounded gains are measured in single percentage points, not tens.

Utah is the centerpiece, but the paper’s claims are about tracking and persistence, not about one state. I assemble parallel administrative panels for five additional interscholastic leagues—Colorado, Minnesota, Wisconsin, Virginia, and Alabama, 47,000 panel rider-seasons beyond Utah’s 29,000—and run each through the same entity resolution, censoring rules, and estimators. Three portable facts result. The persistence-by-rank gradient appears in every league, including untracked Colorado, which is why descriptive gradients cannot carry causal weight. The causal price of a top-five finish is positive and not statistically distinguishable across leagues (pooled, +2.7 percentage points per finish; excluding Utah, +2.3 and not significant)—precisely measured in Utah, the largest panel with the strongest first stages and the only league examined whose divisions are assigned by administrative rule rather than chosen, and individually underpowered elsewhere. What differs across leagues is the *supply* of recognition that structure creates relative to each league’s own merged-field benchmark: Utah’s divisions award 2.35 times the top-five finishes that merged fields would, Minnesota’s and Wisconsin’s partial structures 1.9–2.2 times, Colorado’s undivided fields 1.04 times. At a roughly constant price, leagues that subdivide mint more of what keeps riders racing—as a partial-equilibrium accounting at a constant per-finish price, worth roughly 0.3 to 1.2 percentage points of aggregate annual persistence in the divisioned leagues, and approximately nothing in Colorado. That accounting is an upper bound: the paper’s own dilution estimate implies the per-finish price falls as recognition becomes more common, cutting the constant-price contribution substantially—to near zero if the dilution schedule is taken at face value.

The paper contributes to three literatures. To the economics of tracking and ability

grouping (Betts, 2011; Duflo, Dupas and Kremer, 2011; Card and Giuliano, 2016; Pop-Eleches and Urquiola, 2013; Abdulkadiroğlu, Angrist and Pathak, 2014), it offers the cleanest available estimate of the local-rank-and-recognition channel, on a participation margin that school settings cannot observe, together with actual design counterfactuals—how many groups, how deep the honors—rather than a binary tracked/untracked comparison. To the ordinal-rank literature (Murphy and Weinhardt, 2020; Elsner and Isphording, 2017; Elsner, Isphording and Zölitz, 2021; Denning, Murphy and Weinhardt, 2023), which established that rank affects investment and outcomes conditional on a given partition, it adds the planner’s problem: if rank matters, the partition itself is a policy instrument, and I estimate and simulate its effects. To contest design and the economics of awards (Lazear and Rosen, 1981; Moldovanu and Sela, 2001; Frey and Gallus, 2017; Tran and Zeckhauser, 2012), it contributes field evidence on how prize structure and segmentation affect a dynamic behavioral margin—continued participation—rather than contemporaneous effort, including direct evidence on recognition dilution.

The paper proceeds as follows. Section 2 describes the institutions, including what recognition consists of and how the placement table works. Section 3 sets out a minimal dynamic framework that defines option value and the planner’s problem. Section 4 describes data, linkage, and measurement. Section 5 establishes the descriptive dynamics and the option-value facts. Section 6 presents the three identification strategies and their validation. Section 7 reports causal estimates and the mechanism evidence. Section 8 runs the same designs in five additional leagues to test portability and to quantify how much recognition each league’s structure creates relative to its own merged-field benchmark. Section 9 evaluates counterfactual designs, annually and then dynamically over full careers. Section 10 concludes. A descriptive comparison to an untracked sport (alpine ski racing) appears in Appendix J.

2 Setting and Institutions

2.1 The league

The Utah High School Cycling League (UHSCL), a member of the National Interscholastic Cycling Association (NICA), grew from roughly 500 riders at its founding to over 3,500 by 2024, making it one of the largest youth cycling programs in the country. Races are held on fall weekends: four to five regional races followed by a State Championship. Utah is divided into geographic regions (four in 2018–2021, six from 2022), and riders compete in the region of their school. I exclude State Championship events throughout, which pool regions and use a different bracket structure—consistent with the league’s own placement formula, which also excludes the championship. Riders re-register each fall; there is no obligation to return, and roughly a third of riders in a typical season do not. That re-enrollment choice is the central outcome of this paper.

2.2 What recognition consists of

Education readers should picture the analogue of honor-roll assemblies rather than professional podiums. At each regional race, the top five finishers in each division are called forward at the post-race awards ceremony, in front of the assembled teams and families; each receives a medal, and their names and teams are announced. Results are posted publicly and accumulate into season series standings, which carry their own end-of-season recognition. For a teenager, a top-five finish is a visible, named, repeated public event witnessed by their school community—arguably closer in social content to making the honor roll than to winning a professional race. The fifth-place rider in a division of forty receives this recognition; the sixth-place rider, seconds behind, does not. Recognition is therefore discrete, local (defined within the division, not the league), and designable: its depth (top five) and its scope (the number and size of divisions) are league choices.

2.3 Categories, divisions, and the placement table

Riders are classified into levels (Middle School, Freshman, Sophomore, JV, Varsity) and, within the Freshman and JV levels, into ability divisions (A through E, strongest to weakest); throughout, *category* refers to the grade-gender levels, *division* to the ability subgroups within a category, and *bracket* to the raw start-group label in the results data (a division where divisions exist, otherwise the category). The Category Placement Table (CPT) assigns riders *within region and year* by, in order: previous-year category, overall series points, and average lap-time rank over their best three races. The league sets target division sizes based on projected region size, then fills divisions in CPT-rank order until each target is met.

A worked example makes the cascade concrete. Suppose Region 2 JV Boys has 160 riders and the league’s target division size is 40. The CPT first orders the 160: returning JV A riders ahead of returning JV B riders, and so on by previous category; within a previous category, by last season’s series points; among riders without points (or to break ties), by lap-time rank. The first 40 riders in this order become JV A, riders 41–80 become JV B, and so on. The rider ranked 40th and the rider ranked 41st may be separated by a handful of series points—or by nothing more than a tie-break—yet one will spend the season as the weakest rider in a strong field and the other as the strongest rider in a weaker field, with correspondingly different odds of ever being called to a podium. The cutoffs at 40, 80, and 120 are the “Maimonides” discontinuities exploited by the third research design (Angrist and Lavy, 1999).

Two further institutional features matter. First, the ranking is a strong *ratchet*: returning Varsity riders stay Varsity, and downgrades are granted only for medical reasons or into the non-competitive Single-Lap category, so riders effectively cannot be placed below their prior category. Second, riders may petition to change category, but only through the third race of the season; within-season division changes are rare, and I exclude post-switch observations from the analysis.

2.4 The slack in assignment

The CPT is informed by ability but is far from a deterministic ability sort. Regressing division rank (A=1 through E=5) on lagged speed, year effects, and demographics yields $R^2 \approx 0.47\text{--}0.49$ within the most finely sorted brackets: roughly half of division assignment is unexplained by lagged skill. The residual reflects coach discretion (coaches frequently petition to keep teammates together), team composition, rider preferences, imperfect information about off-season development, and the mechanical interaction of the cascade with region size. Figure 1 shows the consequence. The left panel plots the distribution of within-season skill across JV divisions: the divisions overlap substantially, so riders of similar ability routinely compete in different divisions. The right panel shows what that implies: at the same skill level, riders in different divisions face sharply different probabilities of a top-five finish. This overlap is the identifying variation for everything that follows.

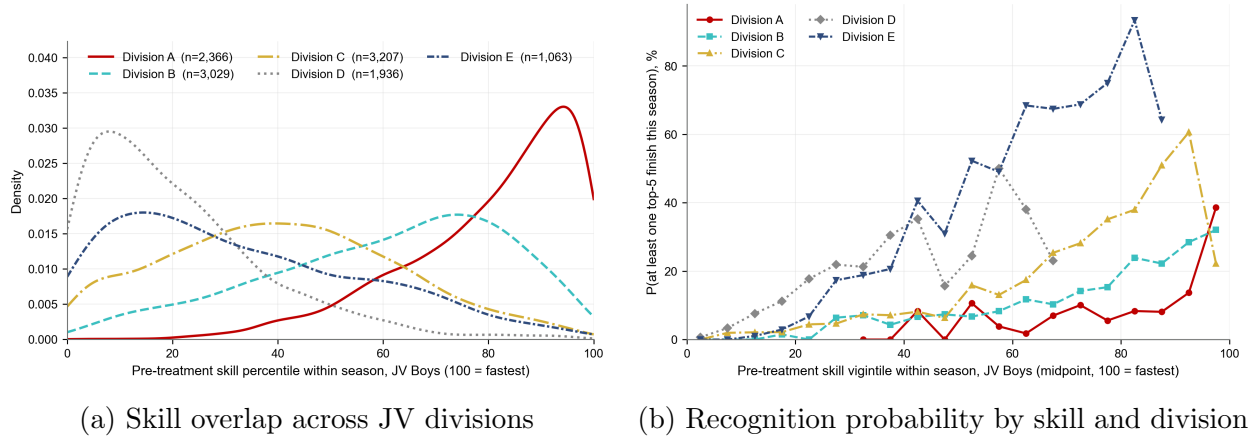


Figure 1: Identifying variation. Riders of similar ability compete in different divisions (left) and therefore face substantially different recognition probabilities (right).

Notes: Within-season skill percentile among JV Boys. Substantial overlap implies that division assignment is not a strict ability sort. Right: share of riders with at least one top-five finish in the season, by within-season skill vigintile and division—at a fixed skill level, recognition odds differ sharply across divisions.

3 Conceptual Framework

A deliberately minimal model fixes the objects the empirics measure. A rider enters season s with ability A_{is} and an idiosyncratic taste for the activity, is assigned to a local field (division), and experiences two kinds of competitive outcomes: a recognition count m_{is} (the number of races in which they finish in the recognized set, e.g. top five) and a continuous local standing r_{is} (the average fraction of their field they beat). The rider returns next season if the utility of participation, including the continuation value of remaining in the activity, exceeds an outside option. Sections 6 and 7 estimate the causal participation responses $\partial P/\partial m$ and $\partial P/\partial r$ using variation in (m, r) generated by quasi-random field assignment rather than by ability, identified locally for each design’s complier population (Appendix Table A7). Ability evolves noisily and exit is effectively absorbing: a rider who does not re-enroll rarely returns. Together these facts generate *option value*—when early signals are noisy, the probability that a currently unremarkable rider ends up varsity-caliber is bounded well away from zero, so retaining them preserves a real claim on future talent that exit extinguishes. Section 5 measures the noise directly and shows the claim is quantitatively large: about a third of eventual varsity seniors were not early standouts.

The league chooses a design $d = (K, k)$: the number (and hence size) of divisions K and the recognition depth k (podium places per division). Two properties of that choice organize the policy analysis (Section 9). **Recognition counts are manufacturable**: total recognition per race is $K \times k$, so more or smaller divisions, or deeper podiums, mechanically create more recognition events, and if utility responds to m the planner can raise aggregate persistence. **Standing is (nearly) zero-sum**: someone must be beaten for someone to beat, so no design can raise *average* standing, and if utility responds only to r , design choices redistribute persistence across ability levels but cannot create it in aggregate. Which mapping operates is therefore not a detail but *the* design question; remaining agnostic, the paper evaluates every policy under both mappings. The planner’s problem, taken up in Section 9, is to choose d to maximize the flow of participants who both survive to their

senior year and clear the top-track (varsity) bar.

4 Data and Measurement

4.1 Sources and linkage

I parse administrative race results for all UHSCL events from 2018 through 2025 from the league’s posted result sheets: 148,876 race-result rows in total. Riders are linked across races and years with a multi-stage entity-resolution procedure (Appendix A) that infers each rider’s grade from bracket and team information, computes an expected graduation year, and splits same-name riders using graduation-year conflicts, school differences, and career-timeline gaps. The procedure resolves 17,729 riders. Five additional leagues—Colorado, Minnesota, Wisconsin, Virginia, and Alabama, about 47,000 panel rider-seasons—are processed through the same pipeline; Section 8 describes them, and one descriptive figure (Figure 2) draws on them earlier.

Because “did not race next season” is itself a linkage outcome—a missed match is observationally a dropout—measurement quality is not a technicality here, and I audit it directly (Appendix A). Three results summarize the audit. First, the resolved identifiers contain zero graduation-year conflicts, and only 2.1 percent of riders are associated with more than two schools. Second, among riders who appear to drop out before their senior year, I search for “near-miss” records (same school and gender, similar name) that could represent failed matches; at most 1.2 percent of apparent dropouts have such a candidate, bounding the scope for false dropouts. Third, the headline return rate is nearly invariant to the linkage regime: 71.9 percent under loose (name-and-gender) matching, 71.7 percent under the baseline procedure, 64.2 percent under strict (name, gender, and school) matching, which mechanically breaks links for school-switchers. Finally, match difficulty is balanced with respect to the identifying variation: riders with common surnames (2.8 percent of the sample) have instrument values indistinguishable from others ($t = 0.1$). Out-of-state moves, injuries,

and switches to other sports remain observationally equivalent to quitting; the estimates should be read as effects on continued participation *in this league*.

4.2 Sample construction

Table 1 reconciles every sample in the paper in one place. From the 148,876 raw rows, entity resolution and cleaning yield a panel of 140,490 race-level observations; excluding State Championships and incomplete results leaves 120,996 rows, which collapse to 38,654 rider-seasons. Excluding the 2025 cohort (whose 2026 return is unobserved) leaves 32,187 rider-seasons of 17,221 unique riders. Because seniors mechanically cannot return to a high-school league, grade-12 rider-seasons are excluded from all persistence regressions, leaving the primary persistence sample of 28,470 rider-seasons (28,434 in the estimating samples, after dropping rider-seasons with missing instrument or control values). The cascade design requires CPT-eligible division brackets and prior-year inputs, so it applies to 17,164 rider-seasons (14,958 after the grade-12 exclusion)—returning riders, roughly half the sample, a fact that matters for interpreting it. The long-run cohort comprises 9,263 riders with an observable senior year (graduation by 2025).

4.3 Treatments and outcomes

The primary treatment is the rider’s seasonal recognition count: the number of top-five finishes, expressed throughout on an attendance-standardized basis (the per-race top-five rate multiplied by the median season length of three races), so that the treatment does not mechanically embed how many races a rider chose to attend; the number of races entered is separately controlled everywhere (Section 6 explains why this matters). As a continuous alternative, I use the season-average fraction of the field beaten. The primary outcome is an indicator for any race appearance the following fall. Long-run outcomes include ever racing as a senior, racing *varsity* as a senior (the policy-relevant top-track outcome), and a stricter top-decile-ability benchmark.

Table 1: Sample accounting

Step	N
Raw parsed race-result rows (all PDFs)	148,876
Rider-resolved panel (v2 identifiers)	143,186
Clean analysis panel (non-competitive categories dropped)	140,490
excluding State Championship races	125,530
finished with valid time and placement	120,996
Rider-seasons	38,654
excluding 2025 cohort (next season unobserved)	32,187
excluding grade 12 (persistence sample)	28,470
Cascade-eligible (division brackets with CPT pools)	17,164
cascade persistence sample (non-senior)	14,958
Unique riders in analysis sample	17,221
Long-run cohort (observable senior year, graduation ≤ 2025)	9,263

Notes: Cascade rows are subsets of the excluding-2025 sample. The grade-12 exclusion reflects that seniors mechanically cannot return to a high-school league; including them would attenuate every persistence estimate with structural zeros. Estimating samples drop a further handful of rider-seasons with missing instrument or control values (e.g., $N = 28,434$ in Table 3).

4.4 Pre-treatment ability

Pre-treatment ability is the course-adjusted speed residual from a regression of log average lap time on venue, year, category, and laps ($R^2 = 0.41$), measured using only prior-season information and shrunk toward grade $\times$ gender $\times$ year cell means by empirical Bayes (Appendix B). The measure is highly stable (season-to-season autocorrelation 0.82), which validates it as a skill measure and simultaneously makes the unexplained half of division assignment more striking. For riders in their first season—who have no prior season—ability and expected-recognition measures used in the long-run analyses are built from the *first race only*, with outcomes defined over subsequent races, so that the right-hand side is predetermined with respect to everything it is asked to explain (Section 6 and Appendix B).

5 Descriptive Dynamics and Option Value

5.1 The motivating fact: local standing tracks persistence within every group

The pattern that motivates the paper is visible in the raw data of every league I observe. Figure 2 plots, for Utah and the five comparison leagues introduced in Section 8, the share of riders who return the following season against their finishing decile *within their own category*. The gradient repeats in every league and every category block: riders in the bottom decile of their category return at rates 15 to 52 percentage points below riders in the top decile. And at every dashed category boundary the rate resets—the sawtooth: riders at the top of one category out-return riders at the bottom of the category above, even though the latter are faster. The same pattern holds within Utah’s ability divisions after adjusting for pre-treatment ability and grade (Appendix Figure A1), so this is not merely that faster or older riders persist: it is *local* standing, conditional on ability, that tracks continued participation. A rider of given speed placed where that speed ranks well looks more likely to stay than the same rider placed where it ranks poorly. The gradient alone cannot distinguish skill sorting from a causal effect of competitive success—it is just as steep in Colorado, which runs no ability divisions, as in Utah. The rest of the paper asks whether the relationship is causal and what it implies for design. Appendix C collects this section’s supporting exhibits.

A two-way cut sharpens the fact and previews the inference problem. Splitting Utah riders by global ability quintile and by their local experience (Appendix Figure A2), riders who collected at least one top-five finish return at 83–90 percent across the *entire* ability distribution—a bottom-quintile rider with a podium out-returns a top-quintile rider who finished mid-pack—while uncelebrated mid-pack riders display the ordinary ability gradient (67 to 77 percent). Most of that boost attaches to *reaching the front of a group*, not to the recognition event itself, and who reaches the front of a group at a given ability is not random. The designs in Section 6 are built to isolate the component that the recognition



Figure 2: The persistence-by-rank sawtooth in all six leagues

Notes: Next-season return rate by within-category finish decile (1 = worst, 10 = best), pooled across genders, by league and category block (Middle School, Freshman, Sophomore, JV, Varsity; blocks with at least 150 rider-seasons shown, so not every league displays every block), with categories adjacent on a continuous axis; dashed verticals mark category boundaries; grade-12 rider-seasons excluded throughout. League panels: Utah 29,101 rider-seasons (2018–2025), Colorado 16,341 (2012–2025), Minnesota 16,500 (2015–2025), Wisconsin 7,575 (2016–2025), Virginia 4,060 (2016–2025), Alabama 2,742 (2021–2026). Top-minus-bottom-decile gaps range from 15 to 52 percentage points and appear in every league, including untracked Colorado—the descriptive gradient conflates skill sorting with causal effects of success, motivating the instrumental-variable designs.

event itself causes.

5.2 Early ability badly under-predicts the top track

A second descriptive fact gives the design question its force: early performance imperfectly predicts who eventually reaches the top track. Appendix Figure A3 plots the probability of racing varsity as a senior against a rider’s early-years (grades 7–9) performance decile. The gradient is real but far from deterministic: 43 percent of the fastest early decile race varsity as seniors, but so do meaningful shares well down the distribution. Read in reverse, this is the option-value fact: **about one-third of riders who race varsity as seniors came from outside the top three deciles of early performance**, and one in seven came from the bottom half. The transition matrix from initial decile to senior outcome shows mass spread far from the diagonal, and survival in the sport, while higher for stronger entrants, is substantial across the entire ability distribution (Appendix Figure A4). In the language of the framework, the innovation variance in ability is large relative to its cross-sectional spread: the league cannot tell at entry who will be varsity-caliber at graduation, so every early exit destroys a real option.

5.3 Early rank luck and the long-run pipeline

Does early local success—as opposed to early ability—reshape the pipeline? I regress riders’ long-run outcomes on the expected recognition implied by their *very first race*: the expected per-race top-five probability computed from the rider’s race-one performance and the composition of the field they happened to face, conditioning on initial-ability decile and graduation-cohort fixed effects. Because the measure uses only race one, with outcomes defined afterward, it is predetermined with respect to all subsequent feedback (the timing issue this construction resolves, and the sensitivity of earlier variants to it, are detailed in Appendix B). Table 2 reports the results. Early rank luck raises eventual *varsity* participation strongly and robustly: +32 percentage points per unit of expected race-one top-five

probability ($t = 7.2$), or +15 percentage points per expected top-five count ($t = 8.6$). The estimate is stable under much richer ability controls—+30 with twenty ventile fixed effects and +30 adding a quartic in the race-one performance percentile ($t \approx 6.7$ throughout)—so the result is not an artifact of coarse race-one ability adjustment. Effects on the top-talent and varsity-senior outcomes are similar. By contrast, under this strictest timing the effect on cumulative seasons raced is a null, and the effect on merely racing as a senior is positive but imprecise ($t = 1.6$): what early success appears to change is not the total amount of racing so much as whether a rider stays on the track that leads to varsity.

Table 2: Long-run reduced forms: race-one rank luck and the pipeline

Outcome	Per unit of race-one top-five rate		Per expected top-five count	
	Coefficient	t	Coefficient	t
Future seasons raced (count)	+0.06	0.5	−0.01	−0.3
Ever races as senior (pp)	+7.3	1.6	+3.1	1.7
Ever races varsity (pp)	+32.0	7.2	+14.8	8.6
Becomes top talent (pp)	+30.7	9.0	+12.2	8.6
Varsity/elite as senior (pp)	+21.4	6.2	+10.1	7.1

Notes: Each cell is from a rider-level regression of the column outcome on first-race expected recognition, conditioning on initial-ability decile and graduation-cohort fixed effects, with region×entry-year fixed effects. Expected recognition uses only race-one information; outcomes are defined over subsequent races. Non-returns are coded zero for senior/varsity outcomes. SEs clustered by rider. $N = 8,458$. Appendix B reports the original (pre-correction) variant for comparison.

5.4 Who is moved, and onto which margin

Appendix Table A3 splits the race-one reduced form by the rider’s initial-ability tier and reports effects *per standard deviation of the instrument within the tier*, alongside each tier’s instrument distribution and baseline outcome—the presentation a skeptical reader needs, because the instrument’s mass differs enormously across tiers (mean expected top-five count 0.60 for elite entrants versus 0.02 for the bottom 40 percent); Appendix Figure A5 shows the corresponding binned reduced forms. Two results organize the design analysis. First, the varsity-margin effect is positive in *every* tier: +9.2 percentage points per SD for elite

entrants ($t = 7.7$), +2.4 for middle entrants ($t = 3.0$), and +2.1 for bottom-tier entrants ($t = 1.9$). Second, the top-decile-*ability* effect exists only for elite entrants (+7.2 per SD, $t = 8.8$) and is essentially zero for everyone else. Early local success moves ordinary riders *into the varsity category*; it does not turn them into top-decile athletes. This asymmetry is the bridge between the causal estimates and the policy analysis: the talent the design margin produces is varsity-level participation, not elite ability, and I hold the simulation in Section 9.2 to exactly that standard.

6 Identification

The correlation between competitive success and persistence conflates ability, unobserved commitment, and the causal effect of success itself. I isolate the last channel with three complementary designs, all built on the same primitive: a rider’s expected competitive success depends on the field they are assigned to, which is only loosely tied to their own ability—and the field they are assigned to moves, year to year, for reasons outside their control. Figure 3 shows the mechanical component of that variation, using the placement queues of one region’s JV Boys category in two adjacent years. A rider’s predicted standing within their assigned division is a sawtooth in queue position; between 2022 and 2023 the division cutlines in this pool moved, and the shaded 90 of 334 comparable queue positions—27 percent—changed division letter outright: the same position that put a rider at the top of division B in 2022 put them at the bottom of division A in 2023. The featured pool sits in the upper quartile of a broad distribution: across all 71 region-year pool pairs, the median share of queue positions that flip division is 20 percent (interquartile range 7–27), and 61 percent of the riders who appear in the same category in consecutive years draw a different predicted division than the year before. Nothing about a rider’s own ability need change for their local rank, and with it their recognition chances, to be remade by enrollment and composition shifts around them. The cutlines drawn here are the rule-predicted ones and

are only the mechanical part of the story—the written rule reproduces 57 percent of actual assignments (Design 3 below)—so the external and simulation designs additionally exploit the discretionary residual around this cascade. I present the three designs in order of the cleanliness of their exclusion restrictions, not the size of their first-stage F -statistics.

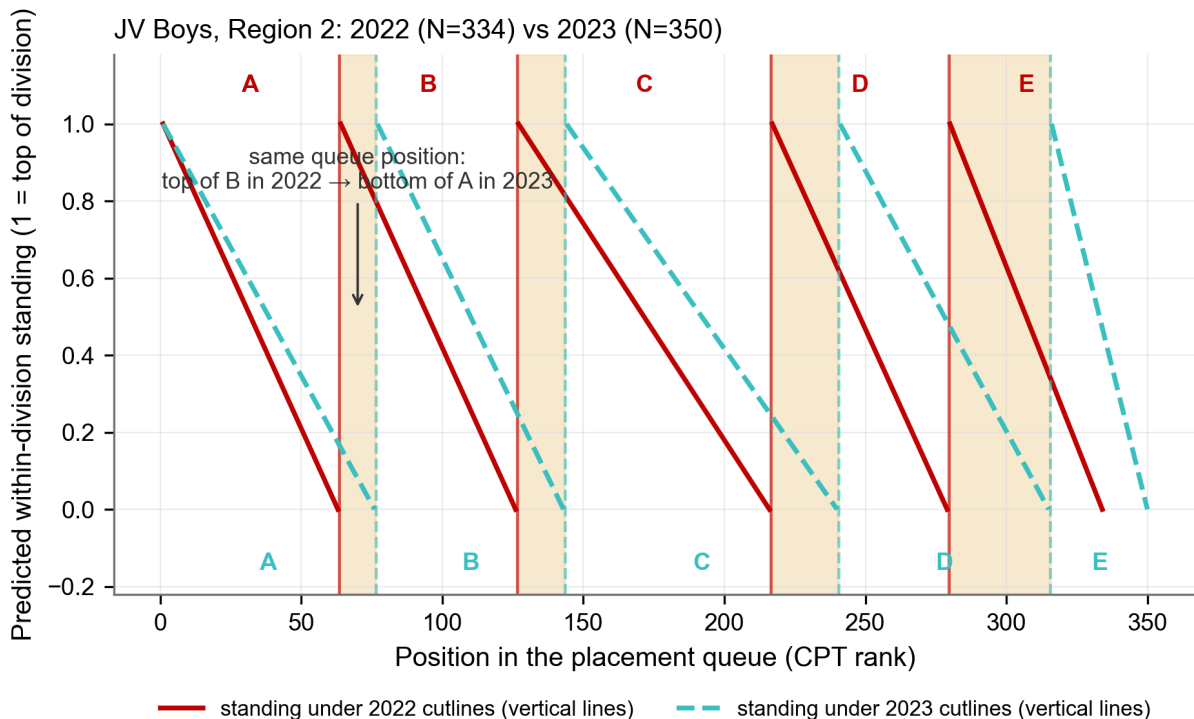


Figure 3: The moving sawtooth: cutlines shift with the field

Notes: JV Boys, Region 2, placement queues of 2022 (solid) and 2023 (dashed). Sloped lines give predicted within-division standing at each position in the CPT queue under that year’s division cutlines (vertical lines); letters mark the divisions in queue order. Shaded bands are queue positions assigned to different divisions in the two years (90 of 334 comparable positions; across all 71 region-year pool pairs the median flipped share is 20 percent, interquartile range 7–27). Division sizes are the observed ones; the queue is the league’s documented lexicographic placement rule (Section 2).

6.1 A pairwise model of expected outcomes

All three designs use a closed-form pairwise win-probability model: rider j beats rider i with probability $\Phi((A_i - A_j)/(\sigma\sqrt{2}))$, where A is pre-treatment ability and σ the race-day residual SD. For any field, the model returns each rider’s expected rank, win, top-five, and fraction-beaten probabilities (tail probabilities via Monte-Carlo ranking so that exactly k

riders are top- k in every draw). The model predicts realized placement well out of sample (Spearman 0.74; top-five AUC 0.87; Appendix D and Figure A6). The Monte-Carlo draws use empirical residuals rather than normal deviates, so the fat tail of race-day disasters (mechanicals, crashes) is built into the tail probabilities by construction (Appendix D).

6.2 Design 1: External-opponent instrument

The design I regard as cleanest on exclusion grounds splits each rider’s field into own-school and external-school competitors and instruments recognition with the expected recognition implied by the strength of the *external* competitors only, controlling for own-school strength. Conditional on own ability and own-team environment, the external field a rider faces reflects *other coaches’* placement decisions—which riders from other schools were petitioned up, held back, or newly recruited. The baseline construction still evaluates that external field *at* the rider’s own estimated ability—the instrument is a function $g(A_i, \text{external field})$ —so, as in the simulation design, residual own-ability variation could leak through a coarse ability control. A fixed-reference variant closes that door entirely: it evaluates each returning rider’s external field at the leave-one-out median ability of the rider’s $\text{season} \times \text{category} \times \text{lagged-ability-vigintile}$ cell, so no function of the rider’s own estimated speed enters the instrument, and the reference-cell level is absorbed in estimation. “Cleanest” refers to freedom from own-ability contamination: external-field strength still moves continuous standing, race difficulty, and peer environment together with recognition chances, and the horse race and channel decomposition of Section 7 are the tools that separate those bundled channels. Monotonicity is plausible and testable in sign: a stronger external field weakly lowers every rider’s expected recognition (first-stage coefficients are positive for every subgroup; Appendix Table A7), and I find no subgroup with a reversed first stage. The complier population is broad, though the design moves recognition most strongly for returning riders, whose fields are more predictable.

6.3 Design 2: Simulation instrument

Averaging the pairwise model over the rider’s actual assigned field gives expected recognition for each rider-season—an instrument for realized recognition whose variation, conditional on own ability, comes from the composition of the field the rider was assigned to. This design uses all the assignment slack at once and is by far the most precise. It has two vulnerabilities, and both are handled mechanically rather than by assumption.

The first is look-ahead in the simulated race-day shocks. The simulation draws each rider’s counterfactual shocks from their own residual pool; residuals are centered on a career-mean fixed effect, so a pool that spans the full career—or one restricted to seasons up to t but still centered on the career mean—imports a location shift that exists only because the rider kept racing. Restricting pool membership to seasons up to t *and* demeaning each pool within that window removes the leak exactly. The correction is not cosmetic: the estimate falls from +6.8 percentage points per finish under career pools, to +5.0 with restricted membership but career centering, to +1.9 once the pools are demeaned (Table 4 reports the full ladder, holding the specification fixed throughout). The demeaned construction is the paper’s standard, and—notably—it carries a *stronger* first stage than the contaminated versions. For returning riders a stricter variant uses only seasons before t for both membership and centering, so every input to the instrument is fixed before the season being instrumented begins (+2.3; Table 3). First-year riders have no predetermined window—their ability estimate is built from the same season that generates recognition—so they are never pooled into the headline estimate and are instead estimated with the race-one entrant design described below.

The second vulnerability is that the instrument is a function $g(A_i, \text{field})$ of the rider’s own estimated ability, so misspecification of the ability control $f(A)$ could leak own-ability variation into the instrument. Monotonicity is satisfied by construction: holding own ability fixed, a weaker assigned field raises expected recognition for every rider.

The full-sample estimates are stable against exactly this vulnerability. As $f(A)$ is enriched from ability-decile fixed effects to twenty percentile bins to full ability $\times$ grade $\times$ gender

interactions, the coefficient barely moves (+1.9, +1.8, +2.0 percentage points; Appendix Table A13). A split-sample construction—ability estimated from odd-numbered races, the instrument built on even-numbered races, eliminating correlated estimation error by construction—yields +4.0 ($t = 6.3$), attenuated less than the demeaned baseline and qualitatively intact. The strict returner variant is less settled: it holds under twenty percentile bins (+1.3) and under direct conditioning on the rider’s full lagged history (+1.0), but flips sign under the full interaction battery (-2.5 , $t = -1.7$) on the halved returner sample—a fragility I report rather than average away.

The entrant race-one design. For first-year riders the design mirrors the long-run timing construction of Section 5: ability and the expected-recognition instrument are computed from the opening race alone, the treatment is the top-five count accumulated over races two onward (standardized as elsewhere), and the outcome is next-season participation, with race-one ability deciles and the standard controls. Every right-hand-side input is therefore fixed before the races that generate the treatment. The instrument is strong ($F = 1,668$) and the estimate is a null: -1.0 percentage points per finish (s.e. 1.0), with a rate-on-rate variant at -0.4 (s.e. 2.6; Table 3).

6.4 Design 3: The cascade (Maimonides) instrument

Reproducing the CPT, I rank riders within region×year pools by the league’s documented lexicographic rule—previous category first, then prior-season series points (parseable for 97 percent of division-bracket rows), then lagged lap-time ability—and fill the observed division sizes, generating a sawtooth in predicted local position as the CPT rank crosses a division-size cutoff (Angrist and Lavy, 1999). A cubic in the running variable absorbs the smooth ability–rank relationship, so identification comes from the discontinuities alone. Because the CPT requires prior-year inputs, this design applies only to returning riders ($N = 14,958$ in the persistence sample)—roughly half the sample, and a selected half: its compliers are

concentrated among returning riders (first stage 0.29, $t = 7.5$) and the fast initial-ability tercile (0.56, $t = 9.0$), with essentially no first stage among slow-tercile riders (0.08, $t = 2.4$) or girls (0.14, $t = 2.0$; Appendix Table A7). The global first stage is moderate (partial $F \approx 50$) and the running-variable density shows no bunching at cutoffs (Appendix E), but the reduced form is small (-1.3 percentage points), and randomization inference over cutoff placement cannot distinguish it from noise ($p = 0.46$).

A stacked-boundary diagnostic explains why. Restricting to riders within two to eight CPT-rank positions of a division cutoff and stacking all cutoffs with boundary fixed effects and distance trends, crossing the predicted boundary moves neither realized recognition (first-stage t between -0.7 and $+0.5$ across bandwidths) nor persistence. The reason is institutional: implemented with the league’s own inputs—including actual prior-season series points—the CPT rule reproduces only 57 percent of assignments overall and only 42 percent among riders within two positions of a cutoff, with mispredictions overwhelmingly one division in either direction (Appendix Table A4). Petitions, race-day adjustments through the third race, and coach input dominate exactly where the design needs the rule to bind. I therefore present the cascade as a bounding design: its 95 percent Anderson–Rubin set of $[-16, +4]$ percentage points (donut version, excluding riders within two positions of a boundary: $[-32, +9]$) rules out large effects but does not independently confirm the other designs’ estimates. The failure is informative in its own right: it is direct, rule-based evidence that division assignment retains a large discretionary component conditional on ability and history—the slack that the external and simulation designs exploit. Section 7 returns to what the cascade’s complier population implies for reconciling the designs.

The slot ladder, validated. The placement process can nonetheless be modeled directly, at the cost of two identifying assumptions: take the observed number of *returning* riders in each division at $T+1$ as given, and assume vacated upper-division seats are filled by promoting returners in period- T series-points order (the league ranks on points—a function

of placement—rather than race time). This slot-filling model has three testable implications, and the data confirm all three. First, the no-demotion ratchet binds: only 2.3 percent of within-bracket stayers move down a division. Second, promotions follow points order: among promotion candidates, the rank correlation between period- T points rank and actual promotion is -0.64 and is correctly signed in every cell where it can be computed. Third, divisions do not expand to absorb incumbent surges: regressing upper-division capacity on incumbent supply yields -0.46 rather than the 1.0 of full absorption, so a surge of protected incumbents must crowd marginal riders down the ladder.

Graduation slack and the crowd-out reduced form. The supply of protected incumbents above the B/C margin moves across years for a mechanical demographic reason: how many of last year’s upper-division riders were seniors and graduated (mean senior share 0.15, standard deviation 0.12 within region–bracket pools). Riders who enter the bracket, or sit below the margin, in a high-graduation year face more open seats above them. The reduced form of this graduation slack on the exposed population’s persistence is positive and significant: +87 percentage points per unit of senior share—about +10 percentage points per standard deviation ($t = 2.7$)—with the instrument balanced on exposed riders’ ability ($t = -0.2$); a mild shift toward first-year riders in high-slack years ($t = 1.8$) is controlled throughout (Appendix Table A6).

A predetermined opponent-supply instrument. A natural referee proposal pushes this logic to its limit: instrument a rider’s recognition chances with the predetermined component of opponent supply alone—the composition of prospective competitors implied by last season’s rosters and graduations—discarding every input that reflects current-season choices. Implemented, the design delivers the balance it promises: the instrument is unrelated to the rider’s own lagged ability ($t = 0.7$), lagged top-five count ($t = 0.1$), and lagged standing ($t = 0.2$), with only a modest correlation with lagged races entered ($t = -2.4$). But it has no first stage: the effect on the realized top-five count is a null for all returners ($t = -0.6$) and

for the fastest tercile ($t = -1.0$), so the resulting 2SLS is uninformative (first-stage F below one). The verdict matches the promotion-margin arithmetic below: slices of the assignment process that are fully predetermined are too thin to move recognition detectably, and only the league’s pre-petition placement lists could change that.

A promotion-margin experiment. The slot model also produces, for every returning rider below the top division, a predicted promotion cutoff: the number of seats opened above, filled in points order. Comparing riders just inside versus just outside the cutoff (96 pool-year-division cells; 3,669 returner transitions; cell fixed effects and distance trends), crossing the predicted cutoff raises the probability of actual promotion by 11–14 percentage points ($t = 2.0$ – 2.6 across bandwidths), and barely-promoted riders show the expected drop in next-season local standing (-0.03 to -0.04 fraction beaten). The persistence effect at this margin is an honestly bounded zero: $+2.3$ percentage points (s.e. 8.2) within eight ranks of the cutoff, $+5.0$ (s.e. 4.7) using all candidates (Appendix Table A5). This does not contradict the graduation-slack reduced form above: the two estimate different objects on different populations —slack moves assignment, peer composition, and cohort thinness together for *all* exposed riders, while the promotion design isolates the single marginal seat for returners—and the promotion interval is wide enough to contain effects of the size the slack reduced form implies. The power arithmetic is unforgiving: with an 11–14 point first stage, detecting the headline $+1.9$ effect would require resolving a reduced form of about a quarter of a percentage point against a standard error near five. The binding constraint is cutoff noise from petitions and multi-division moves, which only the league’s pre-petition placement lists could remove. The ladder design therefore contributes direct institutional validation—the cascading margin exists, and assignment slack is demonstrably discretionary—and a promotion-margin estimate consistent with, but unable to confirm, the other designs (Appendix E).

6.5 Validation: balance, placebos, and the experience confounder

Appendix G collects the validation evidence as numbered tables; here I summarize the design-relevant facts.

The experience confounder. A methodological caution with relevance beyond this paper: instruments built from *continuous* standing (fraction beaten, expected rank) are mechanically confounded by experience. A fast newcomer dominates an easy field, so expected standing is high precisely for riders who are new—and being new independently predicts attrition. The confounding is severe: regressing prior-season participation (a predetermined variable) on the standing-type instrument with only skill and year controls yields $t = -39.8$; adding pre-treatment experience controls (race count, first-year status) collapses it to $t = -2.8$ (Appendix Table A11). All specifications in this paper therefore include experience controls, and any rank-based design in any setting that mixes newcomers with veterans should expect the same issue.

Attendance inside the count. A seasonal recognition *count* summed over attended races embeds attendance—a commitment proxy—inside the treatment: the same logic as the experience confounder, aimed at the count instruments. I therefore standardize: the treatment and instruments are per-race rates scaled by the median three-race season, with races entered controlled everywhere. Appendix Table A12 shows the headline estimate under the old (raw-count) and new (standardized) definitions (+0.8 versus +1.9 for the simulation design, the raw count now attenuating rather than inflating; the external design is essentially unchanged, +1.6 versus +1.4), and a fully rate-on-rate specification points the same direction with less precision.

Balance and pre-trends. With experience controls in place, prior-season participation shows small negative residual correlations with every instrument (coefficients -0.008 to -0.021 ; t from -0.9 for the external standing instrument to -7.5 for the simulation count)—

precisely estimated but an order of magnitude too small to matter relative to the estimated effects, and of the wrong sign to generate them. Lagged *performance* measures correlate with all instruments conditional on ability deciles, as they must (the instruments are functions of lagged ability): the strict returner instrument, for example, still predicts the lagged top-five count ($t = 13.0$), and joint tests of five predetermined history variables reject for every instrument, including the strict and fixed-reference constructions (Appendix Table A10). Balance on lagged history is therefore not restored by any construction in this paper, and I do not claim it is. The response is twofold: the estimates are stable as $f(A)$ is enriched far beyond deciles (Appendix Table A13), and conditioning *directly* on the full lagged history—prior top-five count, standing, races entered, and participation—leaves the strict returner estimate at +1.0 (s.e. 1.3), sign and magnitude intact at the price of a wider interval. The full balance matrix, including the regression of season- $(s-1)$ participation on season- s instruments that serves as the pre-trends placebo, is Appendix Table A10.

6.6 Estimating equations and inference

The estimating equations, at the rider-season level, are

$$\begin{aligned}\text{Recognition}_{is} &= \pi Z_{is} + f(A_{is}) + X_{is} + \delta_{rs} + u_{is}, \\ \text{RacedNext}_{is} &= \beta \widehat{\text{Recognition}}_{is} + f(A_{is}) + X_{is} + \delta_{rs} + \varepsilon_{is},\end{aligned}$$

where Z_{is} is the excluded instrument, π the first-stage coefficient, β —the object of interest—the causal effect of recognition on next-season participation, $f(A)$ is a flexible ability control, X includes pre-treatment experience (races entered, first-year status) and, for the cascade, the running-variable cubic, δ_{rs} are region $\times$ year fixed effects (r here indexes regions)—the baseline stratification, since divisions are defined within region-years—and u_{is} , ε_{is} are error terms. Recognition_{is} and RacedNext_{is} are the empirical counterparts of the framework’s m and P ; replacing the count with the season-average fraction beaten gives the

counterpart of r (Panel B of Table 3). Because the instruments vary at the field level (division $\times$ region $\times$ season), rider clustering alone risks overstating precision; I report standard errors clustered by rider and, in brackets, by field (436 fields in the full sample) for every estimate. Two further inference caveats are stated rather than hidden. First, the expected-recognition instruments and the ability control are generated regressors (pairwise model plus empirical-Bayes shrinkage); a full-pipeline bootstrap propagating that estimation error has not yet been implemented, so the reported standard errors do not include generated-regressor uncertainty, and the field-clustered statistics should be treated as the primary guard against overstated precision. Second, for the cascade I supplement asymptotic inference with weak-instrument-robust Anderson–Rubin sets and randomization inference over cutoff placement.

7 Causal Estimates and Mechanism Evidence

7.1 Main estimates

Competitive success raises returning riders’ next-season participation in all three designs, by one to two percentage points per top-five finish (Table 3). Three results stand out.

First, the recognition-count estimates cluster between +1 and +2.3 percentage points per additional top-five finish. The simulation design, with demeaned look-ahead-free pools, yields +1.9 ($t = 3.2$ by rider; $t = 2.5$ by field); its strictly predetermined returner variant yields +2.3 ($t = 2.1$), or +1.2 (s.e. 1.3) when the treatment is scaled per median season rather than per count. The external-opponent design, whose exclusion restriction I regard as cleanest, yields +1.4 ($t = 2.0$ by rider; $t = 1.4$ by field), and its fixed-reference variant—which admits no function of the rider’s own estimated speed—yields +1.5 (s.e. 1.3) with ability-decile controls and -1.4 (s.e. 2.3) when the reference-cell level is absorbed, imprecise in both directions. The cascade’s point estimate is an uninformative +1.3 with an Anderson–Rubin set of $[-16, +4]$. Table 4 shows what the look-ahead correction did to the simulation design: the same specification reads +6.8 under career pools, +5.0 under restricted-membership

Table 3: Effect of competitive success on next-season participation

Design	OLS	2SLS	(rider SE) [field SE]	First-stage F	N
<i>Panel A. Recognition count (pp per additional top-five finish)</i>					
External opponent	+2.6	+1.4	(0.7) [1.0]	1,987	28,434
Simulation (demeaned pools)	+2.5	+1.9	(0.6) [0.7]	3,985	28,434
Strict returner: count	+3.6	+2.3	(1.1) [1.2]	758	13,938
Strict returner: per-median-season	+5.4	+1.2	(1.3) [1.3]	836	13,938
Fixed-reference external	+5.4	+1.5	(1.3) [1.3]	773	13,911
Entrant race-one (races 2+)	+1.7	−1.0	(1.0) [1.7]	1,668	13,067
Cascade (CPT)	+3.5	+1.3	(5.0) [4.7]	50	14,958
AR 95% set	[−16, +4] pp; donut [−32, +9] pp; randomization inference $p = 0.46$				
<i>Panel B. Continuous standing (pp per 0→1 fraction of field beaten)</i>					
External opponent	+25.6	+0.5	(2.6) [3.5]	6,530	28,434
Simulation (demeaned pools)	+22.1	+7.8	(1.6) [2.5]	21,107	28,434
External, recentered	+29.6	−11.8	(5.7) [6.8]	835	14,954
Strict returner	+33.6	−5.7	(3.0) [3.1]	3,514	13,938
Cascade (CPT)	+22.8	−15.2	(20.1) [21.9]	58	14,958

Notes: Outcome is racing the next season; persistence sample (grade 12 excluded). Panel A treatment: one additional top-five finish, standardized to a median three-race season (attendance-purged; Appendix Table A12). Panel B treatment: moving from beating none to all of one’s field. All specifications include region×year fixed effects, ability-decile fixed effects, and pre-treatment experience controls (races entered, first-year status); the cascade adds a cubic in the CPT running variable. Simulation rows use the demeaned, look-ahead-free residual pools (Table 4). “Strict returner” restricts both the ability estimate and the shock pools to seasons strictly before t , demeaned within that window (returning riders only); the count row adds the attendance control, the per-median-season row scales the per-race rate to the median three-race season. “Fixed-reference external” evaluates the external field at the leave-one-out cell-median ability (Section 6), with ability-decile controls. “Entrant race-one” instruments first-year riders’ recognition over races two onward with expected recognition built from the opening race alone. “Recentered” subtracts the rider’s own-ability component from the external standing instrument. SEs clustered by rider in parentheses and by field (division×region×season; 436 fields in the full sample, 373–376 in the returner samples, 342 in the entrant sample, 278–280 in the cascade-eligible samples) in brackets. The 2SLS column reports the coefficient β from the estimating equations of Section 6. The cascade is underpowered; its Anderson–Rubin set and randomization-inference p -value are the appropriate summaries. Reported SEs do not include generated-regressor uncertainty (Section 6).

pools that retain career centering (the pre-revision headline), and +1.9 once the pools are demeaned—the paper’s standard.

Table 4: The look-ahead correction ladder

Residual-pool construction	2SLS	(rider SE) [field SE]	First-stage F	N
Career pools, career-centered (original)	+6.8	(0.6) [1.0]	3,325	28,434
$\leq t$ pools, career-centered (pre-revision headline)	+5.0	(0.6) [0.9]	2,577	28,434
$\leq t$ pools, demeaned (revised standard)	+1.9	(0.6) [0.7]	3,985	28,434
$\leq t - 1$ strict, demeaned (returners only)	+2.3	(1.1) [1.2]	758	13,938

Notes: Effect of one additional top-five finish on next-season participation (pp), simulation design. Every row holds the specification of Table 3 fixed—same treatment, controls, fixed effects, and sample—and changes only how the instrument’s residual pools are built. Row 1 draws each rider’s simulated race-day shocks from residuals spanning the full career, centered on the career mean; row 2 restricts pool membership to seasons up to t but retains career centering, so a location shift that exists only because the rider kept racing survives; row 3 demeans each pool within the $\leq t$ window, removing the leak; row 4 additionally restricts both the ability estimate and the pools to seasons strictly before t (feasible only for returning riders). SEs clustered by rider in parentheses and by field in brackets.

Second, the effect is a returning-rider phenomenon. Re-estimating the two main designs on the cascade-eligible subsample (returning riders in division brackets) yields +2.2 ($t = 2.4$) for the simulation design and +0.9 ($t = 0.8$) for the external design; the cascade’s wide set cannot adjudicate between them. The heterogeneity table (Appendix Table A8) shows every design positive for returning riders—+1.1 (simulation), +2.5 (external), +2.3 (strict count)—while the first-year margin, estimated on its own strictly predetermined race-one design, is a null: −1.0 percentage points (s.e. 1.0) per finish, with the rate-on-rate variant equally empty (−0.4, s.e. 2.6). I keep the two populations separate rather than averaging them: the supportable statement is a returning-rider effect of roughly +1 to +2.3 percentage points per finish and no detectable entrant response.

Third, the continuous-standing estimates change sign with the construction: +7.8 ($t = 4.8$ rider, 3.1 field) in the full-sample simulation design, a null in the external design (+0.5, $t = 0.2$), but *negative* in the recentered external variant (−11.8, $t = -2.0$), the strict returner design (−5.7, $t = -1.9$), and the underpowered cascade. The count parameterization is the more stable object; the standing parameterization is not robust to how its instrument is built. This previews the mechanism evidence below.

Appendix Table A8 reports the heterogeneity the education audience will ask for first. The full-sample count effects are modest and similar across genders (simulation design: boys +1.7, girls +1.5) and across experience groups they separate as above: returning riders +1.1 to +2.3 across designs, first-year riders a null in the entrant design. The strict returner design is too imprecise to support gender splits (-1.9 for boys, s.e. 2.0; $+0.8$ for girls, s.e. 2.0). Fixed-effects and functional-form robustness are collected in Appendix G: the simulation estimate moves from +1.8 (year fixed effects only) to +1.9 (region $\times$ year, the baseline) to +1.6 under school $\times$ year fixed effects, which compare riders on the same team in the same season (Appendix Table A14); enriching $f(A)$ moves it between +1.8 and +2.0 with the split-sample construction at +4.0 (Appendix Table A13).

7.2 Mechanism evidence: counts, standing, or the medal moment?

Whether the effect is carried by discrete recognition events or by continuous local standing determines which design levers work (Section 3), so I run every test available and report them all, committing to none (Appendix F collects the full exhibits).

The horse race. Table 5 instruments both treatments simultaneously, and the verdict depends on the construction. In the full-sample joint 2SLS, continuous standing carries the effect (+6.7 per $0 \rightarrow 1$, $t = 3.2$) while the top-five count, conditional on standing, is a null (+0.6, $t = 0.9$); the joint reduced form gives the same answer (standing +7.6, $t = 3.6$; count +0.55, $t = 0.9$). The less demanding specifications agree: the count effect conditional on fraction-beaten deciles is *negative* (-5.6 , $t = -3.6$), while standing conditional on top-five-count dummies is positive but insignificant (+2.6, $t = 1.2$). Yet the strictly predetermined returner design of Table 3 delivers exactly the opposite ranking: counts positive (+2.3, $t = 2.1$) and standing negative (-5.7 , $t = -1.9$). The cascade version is uninformative. Identification in the horse race comes from riders with identical standing but different counts (variation in field size and race-day noise), which is genuine but thinner than the

single-treatment variation. I read the evidence as a decomposition that is not robust across instrument constructions: the full sample favors standing, the strictly predetermined returner sample favors counts, and neither verdict survives the other’s construction.

Table 5: Mechanism horse race: recognition count versus continuous standing

Specification	Top-five count (pp)	Fraction beaten (pp)
Joint 2SLS (simulation instruments)	+0.6 (0.7)	+6.7 (2.1)
Joint reduced form	+0.6 (0.6)	+7.6 (2.1)
Top-five fraction-beaten decile FE	−5.6 (1.6)	—
Fraction beaten top-five count FE	—	+2.6 (2.1)
Joint 2SLS (cascade instruments)	+2.5 (5.6)	−22.3 (22.9)

Notes: Outcome is next-season participation; persistence sample ($N = 28,434$; cascade row 14,958). Count effects per additional top-five finish; standing effects per 0→1 fraction beaten; the joint reduced form is per unit of the corresponding expected-recognition instrument. Rider-clustered SEs in parentheses. Controls as in Table 3. Simulation instruments use the demeaned, look-ahead-free pools throughout. In-text t -statistics are computed from unrounded estimates and may differ in the final digit from ratios of the rounded entries shown here.

The medal moment is not the margin. If recognition operates through the discrete event of medaling at a given race, riders finishing fifth versus sixth should differ in persistence. Pooling every fifth/sixth-place comparison across all races and seasons in a local-polynomial design (Appendix Figure A8), the discontinuity is a precise zero: +0.4 percentage points (SE 1.0) at a bandwidth of 3.5 placement positions ($N = 15,619$), with a minimum detectable effect at 80 percent power of 2.5 percentage points—well below the per-finish causal estimate. Estimates at narrower bandwidths are similar (−1.4 and +1.4, both insignificant). Whatever recognition does, it does not do it through the medal ceremony of a single race: the operative treatment is cumulative over the season. Appendix Figure A9 corroborates this nonparametrically: persistence declines smoothly in season-average placement with no discontinuity at the medal cutoff, and placebo gaps between adjacent finishers are null.

Recognition dilutes. If recognition derives part of its value from scarcity, the per-finish effect should decline where a larger share of the field is recognized—and it does. Interacting the (instrumented) top-five count with the field’s demeaned medal share yields a main effect

of +1.6 percentage points ($t = 2.6$) and an interaction of -0.150 ($t = -3.3$; Appendix Table A6): each 10-percentage-point increase in the share of the field recognized reduces the per-finish effect by about 1.5 percentage points, driving it to zero at a recognized share of about 17 percent, against an observed mean of 7 percent. This is the athletic analogue of grade inflation eroding the signal value of honors (Butcher, McEwan and Weerapana, 2014), and I carry it into the policy section as a first-order caveat on recognition-broadening counterfactuals, which extrapolate toward exactly the high-share regimes where the effect is weakest.

Counts versus standing as parameterizations. The pairwise model translates one treatment into the other. At observed field sizes, moving from beating none to all of one’s field corresponds to about 1.8 additional top-five finishes per (three-race) season on average (ranging from 1.5 in fields of 40-plus to 3.5 in fields under 25), so if counts were the structural treatment, the +1.9 count estimate would imply a 0→1 standing effect of about +3.4 percentage points. The directly estimated standing effect is more than twice that (+7.8) in the full sample, and has the opposite sign (-5.7) in the strict returner construction, where the count-implied value would be about +4 (Appendix Figure A10). The two parameterizations therefore do not describe one dose–response: the count parameterization is stable across designs and controls, while the standing parameterization changes magnitude and sign depending on how its instrument is built and what it is conditioned on—evidence that the continuous rank is not a structural treatment in these data, without licensing the converse claim that the discrete event is.

Where the verdict stands. Neither mechanism wins. The recognition-count treatment is the more stable single-treatment object—it holds its magnitude across designs, genders, fixed-effect batteries, and the attendance purge—but it loses the full-sample horse race to standing and turns negative conditional on standing deciles. The standing treatment wins that horse race yet flips sign in the strictly predetermined returner and recentered construc-

tions. The dilution pattern sits more naturally with scarcity stories, and the medal-moment null says the operative object is season-cumulative rather than a single ceremony. No simple version of either mechanism fits all of these facts. The paper’s policy analysis therefore carries both mappings everywhere (Section 9), and readers should hold conclusions that depend on either mechanism exclusively—particularly recognition-depth levers, which lean on the count mapping—more loosely than those that survive both.

Participation, not performance. Recognition could in principle operate on an intensive margin as well: riders who taste success might train harder and ride faster. I find no evidence of this, on either of two margins. First, next-season speed: instrumenting competitive success as in the main design, with the same ability-control battery the persistence results survive (Appendix Table A13), the effect of an additional top-five finish on next-season course-adjusted lap pace is +0.1 percent (s.e. 0.3)—ruling out speed gains larger than roughly one year of typical adolescent development.¹ Second, contemporaneous effort: the same rider, within the same season, rides no faster (if anything marginally slower) in races where the day’s start field places them in podium contention than in races where it does not (within rider×season fixed effects, every expected-rank bin within ± 0.15 percent of the rider’s own average pace), and pace shows no jump at the division-promotion cutoff. The recognition incentive operates on whether riders show up—this season and next—not on how hard they ride once the race starts.

7.3 The channel decomposition: what is “recognition” made of?

The instruments shift division assignment, which moves a bundle: recognition chances, the division label (A versus E—prestige and stigma, the tracking literature’s classic concern), peer quality, and race difficulty. For an education audience this bundle is not a nuisance but

¹Speed outcomes are demanding: pace is nearly collinear with fine-grained ability, so modest ability-control error masquerades as treatment effects (a naive specification yields a spurious +1.6 percent, $t = 5.4$, which the interacted ability controls eliminate), and roughly forty percent of a season-mean pace deviation is transitory and mean-reverts (Appendix Table A15).

the question: *which* component of track placement moves persistence? Appendix Table A9 unbundles it by adding controls that absorb each channel in turn, within the division-bracket subsample where labels exist. One finding is stable: the label itself (being “an A rider” versus “a C rider”) is not what carries the effect. The recognition-count effect is +2.1 in that subsample at baseline ($N = 14,931$, a few rows smaller than the cascade-eligible frame of Table A6, which yields +2.2) and is *larger* within division labels (+4.0, $t = 4.1$), with the external instrument agreeing (+0.9 at baseline, +2.6 within labels). Beyond that, the decomposition loses its footing at the corrected magnitudes. Adding peer-quality controls (division mean and SD ability, own-school teammate count and strength) cuts the estimate to +0.9 ($t = 0.8$), and adding field size takes it to -0.5 ($t = -0.5$); the external version flips to significantly negative under the same battery (-4.2 , $t = -3.2$), while the continuous-standing treatment under the full control set is +12.9 ($t = 3.2$)—sign-opposite to its strict-design estimate. Peer quality and field size are themselves the ingredients from which expected recognition is computed, so the later rungs of the ladder absorb much of the identifying variation along with the channels they name. The supportable conclusions are one positive and one negative: the effect does not run through the track’s name (Frey and Gallus, 2017; Tran and Zeckhauser, 2012), and these data cannot cleanly separate recognition chances from the field composition that produces them—the channel evidence should be weighted accordingly.

8 Evidence Beyond Utah: One Price, Different Supply

The mechanisms studied above are not Utah peculiarities: five additional leagues, processed through an identical pipeline, test whether they travel. Appendix H collects the supporting exhibits. I assemble administrative race results for five additional interscholastic mountain-bike leagues—Colorado (2012–2025), Minnesota (2015–2025), Wisconsin (2016–2025), Virginia (2016–2025), and Alabama (2021–2026)—and process each through the same pipeline as

Utah—the same entity resolution, grade inference and graduation censoring, course-adjusted speed measure, and estimators—so differences in results reflect the leagues, not the methods. Together the six leagues contribute 76,300 rider-seasons (panel totals; estimation samples are smaller because the instruments require a prior-season speed measure), 54,300 of them with an observable next-season outcome. The leagues differ in exactly the dimension the paper is about: Utah subdivides each grade-gender category into ability divisions (2.18 separately awarded divisions per category on average); Minnesota, Wisconsin, Virginia, and Alabama subdivide partially (1.2–1.3); Colorado runs each grade-gender category as a single start (1.00). League-by-league return rates range from 63 to 74 percent.

Three facts emerge, and together they discipline how the Utah results generalize.

The persistence gradient is universal—and not causal by itself. The motivating sawtooth (Figure 2) already pooled these leagues: the within-category gradient and its boundary resets replicate in every league and every category block, and are just as steep in Colorado, where there are no divisions, as in Utah. The gradient alone therefore cannot distinguish skill sorting from a causal effect of competitive success—which is why the paper leans on instruments, and why replicating the *causal* estimate across leagues is the informative exercise.

The per-finish estimate is positive everywhere it is precise, and no league rejects a common price. Table 6 reports, for each league on an identical returning-rider frame, the OLS gradient and the two instrumental-variable designs of Section 6 (the cascade applies only in Utah, below). Pooling all six leagues with league-specific match cells, one additional top-five finish raises next-season participation by 2.7 percentage points ($t = 2.2$). Utah is the only league individually significant (+3.4 to +5.7 across designs; on this trimmed returning-rider frame the external design sits above the simulation design, reversing their full-panel ordering—the two designs weight different complier mixes), and it largely carries the pooled result: excluding Utah the pooled estimate is +2.3 (se = 1.5), positive but not significant.

The summary has two halves. Outside Utah the estimates are individually underpowered: the cross-league exercise cannot independently confirm the causal effect, but every league’s confidence interval is consistent with the pooled price. Utah—the largest panel with the strongest first stages, though on this frame (which requires consecutive-season participation and trims the lagged-speed tails) it is estimated above its own within-paper returning-rider range of +1.1 to +2.5 (Appendix Table A8)—is where the price is measured.

A direct test of heterogeneity cannot reject equality, but only weakly: interacting the instrumented top-five count with an indicator for division-running leagues yields an increment of +0.5 percentage points ($se = 2.3$), a null that is bounded, not tight—differences in the price smaller than roughly 4.5 percentage points, larger than the pooled price itself, would go undetected. One leg of that comparison is also compromised: Colorado, the only fully untracked league, fails the placebo test below.

The placebo and overidentification evidence bears this out. Weak-instrument-robust inference changes nothing because there was nothing fragile to protect: Anderson–Rubin sets are wide for the small leagues (Virginia, Alabama), and wild-cluster bootstrap p -values for Alabama (0.13) and Virginia (0.14) likewise fail to reject—those leagues are individually uninformative, not confirmatory. Placebo regressions of *prior* speed improvement on the current instrument pass in Utah, Wisconsin, and Minnesota ($p = 0.32, 0.95, 0.23$) but fail in Colorado ($p = 0.003$), so Colorado’s imprecise estimates carry a trajectory caveat. Virginia and Alabama have too few consecutive three-season careers to run the test. Overidentification tests using both instruments jointly reject nowhere ($p = 0.11$ – 0.98). A further caution applies to single-start leagues: with no divisions, the simulation instrument’s variation comes from race-to-race attendance and entry composition rather than from administrative assignment—a margin closer to rider choice—so the Colorado and Virginia columns are weaker tests of the causal claim than Utah’s.

Table 6: The per-finish effect on an identical frame, six leagues

League	N	OLS Effect of one additional top-five finish on next-season return (pp)	External-opponent IV (Design 1)	Simulation IV (Design 2)	Assignment-rule IV (Design 3, cascade)
Utah	8,362	+2.8 (0.4)	+5.7 (2.3) [F=361]	+3.4 (1.9) [F=238]	+4.5 (3.8) [F=131]
Colorado	4,303	+1.8 (0.6)	-5.9 (8.9) [F=23]	+4.3 (3.3) [F=77]	—
Minnesota	3,136	+2.6 (0.7)	+2.9 (4.2) [F=112]	-0.1 (2.8) [F=94]	—
Wisconsin	1,278	+0.6 (1.0)	+0.1 (5.0) [F=60]	+0.2 (2.5) [F=113]	—
Virginia	705	+6.8 (1.9)	weak instr. [F<1]	+11.0 (7.1) [F=33]	—
Alabama	374	+2.2 (2.6)	+5.5 (5.8) [F=42]	+9.0 (6.0) [F=40]	—
Pooled (six leagues)	18,158	+2.5 (0.3)	+5.4 (1.9)	+2.7 (1.2)	—
Pooled excl. Utah	9,796	—	—	+2.3 (1.5)	—

Notes: Returning riders with observable outcomes, grade-12 excluded; within-cell demeaning on gender $\times$ category $\times$ season $\times$ skill-ventile cells with a cubic in lagged speed and races entered; standard errors clustered by rider; first-stage F on the row beneath each estimate. “Weak instr.”: point estimates are suppressed wherever the first-stage F is below 10; “—” marks designs that do not exist in a league (the cascade requires a league-administered assignment rule, which only Utah has). N is smaller than the within-Utah returning-rider samples of Section 7 because this frame requires consecutive-season participation for the lagged-speed measure and trims the top and bottom 5 percent of the lagged-speed distribution. Pooled rows stack the six trimmed league samples (so N equals the sum of the league rows) and include league in the match cells. The pooled OLS and external-opponent estimates are from the stacked specification ($N = 18,164$; league-specific cells); the pooled simulation estimate is from the match-cell specification ($N = 18,158$). These are returning-rider estimates on a deliberately identical cross-league frame; Utah’s richer within-league designs (Section 7) remain the paper’s primary estimates.

League-administered assignment exists only in Utah. The cascade logic of Section 6 doubles as a structural diagnostic: if a league fills ability divisions by rule, predicted assignment under perfect rank-based sorting should predict actual assignment. In Utah it does (first-stage coefficient 0.50, $t = 37$). In Wisconsin and Minnesota it does not (coefficients 0.05 and -0.02 , $t < 1$): their JV2/JV3 “tiers” are race-distance choices made by riders and coaches, not league-imposed ability assignments. (Virginia and Alabama subdivide too little to run the diagnostic; the claim is about the leagues examined.) Utah is thus not merely the biggest sample; it is the only league examined whose division structure is an administratively assigned treatment—the variation a study of tracking needs, and the reason it anchors the paper. Two clarifications keep this paragraph consistent with Section 6: the 0.50 coefficient is the assignment-on-predicted-assignment first stage, not the recognition first stage (the corresponding instrument F for the top-five count is 131 in Table 6); and it coexists with the earlier finding that the written rule reproduces only 57 percent of individual assignments—the rule predicts the broad division well and the marginal slot poorly, which is exactly why the cascade bounds rather than pins down the effect.

What structure does is change the supply of recognition. If the per-finish price is roughly constant, division structure matters through arithmetic: subdividing a field of fifty into two fields of twenty-five doubles the number of top-five finishes awarded. Appendix Table A16 quantifies this with a within-league accounting counterfactual: merge each race’s category-gender field into a single start, hold attendance fixed, and count top-five slots. Two different objects must not be conflated here. The *level* of recognition per capita is dominated by league scale and field size, not structure—small-field Virginia awards more top-fives per 100 rider-seasons (57) than Utah (36), and Colorado (32) is within four of Utah. The structural object is each league’s own *multiplier*: actual slots relative to its own merged counterfactual. On that margin Utah’s divisions multiply recognition 2.35-fold (36 top-five finishes per 100 rider-seasons against 15 merged); Minnesota’s and Wisconsin’s

partial structures multiply it 1.9–2.2-fold; Colorado’s single starts multiply it essentially not at all (1.04; the residual 4 percent comes from occasional split and combined-grade starts). Pricing the extra slots at the pooled estimates implies that division structure contributes roughly 0.3 to 1.2 percentage points of aggregate annual persistence in the divisioned leagues and approximately nothing in Colorado, relative to each league’s own merged benchmark. Figure 4 summarizes the section in one picture: no league’s price is distinguishable from the pooled price; the quantity is a policy choice. The incidence follows from the same arithmetic, and Appendix Figure A11 shows it directly: the additional slots accrue to mid-ability riders. A median-ability Utah rider finishes top-five in 42 percent of starts in the lowest division and essentially never in the top division (0 percent); the same median-ability rider in Colorado’s merged starts finishes top-five in under 1 percent of starts. This is the supply-side counterpart of the within-Utah evidence that recognition moves marginal rather than elite riders.

Three caveats bound this accounting. First, it is partial-equilibrium: it holds the price fixed as the quantity changes. The split-sample test of the price in large versus small fields is noisy in both directions and bounds little. More pointedly, the paper’s own dilution estimate (Section 7.2, Table A6) implies the per-finish price *rises* as recognition becomes scarcer, so the merged counterfactual’s price would likely exceed the divisioned price and the constant-price contributions in Appendix Table A16 overstate the structural gain; under the dilution-adjusted pricing used for the Utah counterfactuals in Section 9, the implied contributions shrink by roughly half. Second, the non-Utah contributions price slots at the pooled estimate, which Utah largely carries; the excluded-Utah estimate alone cannot confirm a positive price. Third, the price is identified from returning riders—the population for which all leagues support the same design—so the aggregate contribution inherits that LATE interpretation.

A companion comparison to an untracked *sport*—Utah alpine ski racing, which groups athletes only by age—appears in Appendix J; it does not deliver a clean tracked-versus-

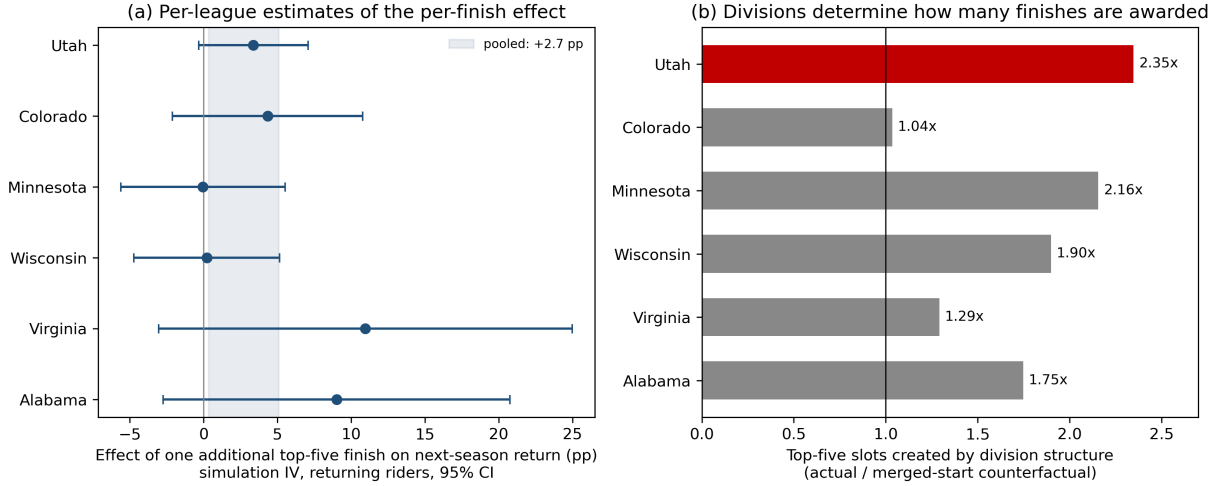


Figure 4: One price, different supply

Notes: Left: simulation-IV estimate of the per-top-five effect on next-season return per league (returning riders, 95 percent confidence intervals, rider-clustered); shaded band is the pooled estimate's 95 percent confidence interval, which is itself dominated by Utah's weight, so the Utah point's agreement with the band is partly mechanical. Right: ratio of actual top-five slots to slots under a merge-all-divisions counterfactual, holding attendance fixed.

untracked contrast and is reported there as descriptive context with its confounds stated.

9 Policy Counterfactuals

Any design induces a pair: aggregate retention and the mean ability of the retained pool. Designs that protect weak performers from low local standing (small groups, broad recognition) raise the first and, mechanically, lower the second—the retention–selection frontier. A planner who cares about the long-run supply of top-track athletes maximizes neither alone but the flow of riders who survive to their senior year *and* clear the varsity bar. This section traces the frontier two ways: an annual exercise that holds each season's rider pool fixed, and a dynamic microsimulation (Section 9.2) that follows entering cohorts over full careers.

9.1 Annual counterfactuals

For each policy I re-assign riders to divisions, recompute every rider’s expected recognition and standing with the pairwise model, and map the changes to persistence using the causal estimates. Counterfactual return probabilities start from cross-fitted baseline propensities estimated on the non-senior population ($N = 28,434$; base return rate 73.7 percent) rather than from realized outcomes, so the mapping never applies the estimated response outside the estimation sample or asymmetrically censors it. Respecting the unresolved mechanism, I price the changes six ways: a *strict* mapping (changes in expected top-five counts priced at the strict returner estimate, +1.2 per median-season finish—the headline pricing), a *count* mapping (the full-sample demeaned estimate, +1.9), an *external* mapping (+1.4), a *joint* mapping using the horse-race coefficients, a *diluted* mapping (the share-dependent count schedule of Section 7.2, floored at zero), and a pure *standing* mapping (changes in expected fraction-beaten priced at the standing estimate). Table 7 reports them side by side.

Table 7: Annual policy counterfactuals under alternative mechanism mappings

Policy	Return-rate change vs. current (pp)						Top-decile talent	
	Strict	Count	External	Joint	Diluted	Standing	Medals per 100	retained (strict)
Current structure	—	—	—	—	—	—	24.9	2,435
One single group	−0.13	−0.20	−0.15	−0.07	−0.23	−0.00	14.4	2,431
One fewer division	−0.06	−0.10	−0.07	−0.04	−0.09	−0.00	19.7	2,431
One more division	+0.15	+0.23	+0.17	+0.08	+0.08	−0.00	37.3	2,436
Top-3 recognition	−0.11	−0.18	−0.13	−0.06	−0.16	−0.00	15.4	2,423
Top-10 recognition ^a	+0.28	+0.43	+0.32	+0.15	+0.09	−0.00	48.3	2,449
Fixed field size 25	+0.53	+0.83	+0.61	+0.28	0.00	−0.00	69.1	2,440

Notes: Each season’s rider pool held fixed; policies re-assign riders and recompute expected recognition and standing with the pairwise model, mapped to return probabilities under the six pricings described in the text (strict returner, full-sample count, external, joint horse-race, dilution schedule, and pure standing). The policy population is the non-senior persistence sample ($N = 28,434$ rider-seasons, grade-12 excluded); counterfactual return probabilities start from cross-fitted baseline propensities on that population (base return rate 73.7 percent), not from realized outcomes. “Top-decile talent retained” counts predicted returners above a fixed top-decile ability threshold (anchored to the current-policy ability distribution, pooled across seasons), under the strict mapping. “Medals per 100” is model-predicted top-five finishes (medal $\equiv$ top-five) per 100 rider-seasons in this sample; it is not comparable to the realized per-start accounting of Appendix Table A16.

^a Out of support: the league recognizes five places, so pricing recognition beyond fifth extrapolates the estimated response outside the observed treatment range.

Three lessons emerge. First, under the count-based mappings, recognition-broadening designs raise persistence: fixing field size at 25 raises the predicted return rate by 0.5 percentage points under the strict pricing (0.8 under the full-sample count pricing) and nearly triples the medal supply; top-ten recognition raises it by 0.3 (0.4); collapsing to a single group lowers it by 0.1 to 0.2. These are small annual numbers, and the dynamic simulation below shows they remain small even after compounding. Second, under the pure standing mapping every aggregate effect collapses to zero (no policy moves the return rate by more than a hundredth of a percentage point), exactly as the framework predicts: standing is zero-sum, so design can only redistribute it. The aggregate case for recognition-broadening design rests on the count mechanism being at least partly operative; the distributional case (moving persistence from riders who would stay anyway to marginal ones) survives either way. Third, the retention–selection frontier is nearly flat: across policies the mean ability of predicted returners barely moves, and the top-decile talent retained varies by at most about one percent. Inclusive recognition buys participation at essentially no cost in the average quality of the retained pool (Figure 5a); the gains from broader recognition are spread across the ability distribution rather than concentrated among the already-recognized, while consolidating into a single group costs most among the faster deciles (Figure 5b).

Two further pricings discipline these numbers, and both are computed rather than conjectured. Priced at the external design’s estimate (+1.4 per finish), the annual gains are +0.32 percentage points for top-ten recognition and +0.61 for fixed-25 fields; priced with the estimated dilution (a share-dependent per-finish effect, floored at zero), they are +0.09 and 0.00—the densest-podium policy loses its entire gain, because it creates exactly the high-share regime in which the estimated per-finish effect vanishes. The policy *ranking* is identical under the strict, count, external, and joint pricings; the dilution pricing demotes fixed-25 fields below top-ten recognition; and under the pure standing mapping every aggregate effect collapses to zero and the ordering weakly reverses (the current structure edges out each broadening policy in Table 7)—the framework’s zero-sum prediction, not a robustness

failure. Across the count-based pricings the magnitudes span from approximately zero (diluted) to +0.8 (full-sample count); I treat the strict returner pricing as the paper’s standard and the full-sample count column as the upper end of the range rather than the headline. All mappings price only the persistence response; entry responses (recognition-rich designs may also attract entrants) are outside the exercise and would push the other way. The full-sample prices are also carried disproportionately by returning riders (Appendix Table A8), so the annual gains inherit the same returner-weighted interpretation as the cross-league supply accounting (Section 8).

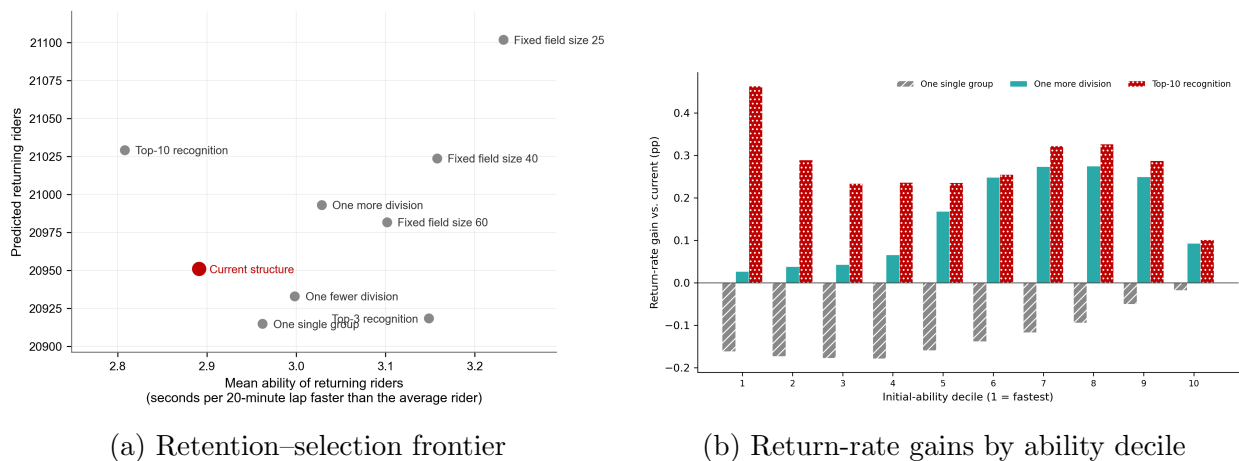


Figure 5: Broader recognition retains more riders at little cost in average ability.

Notes: Annual policy counterfactuals under the strict returner (headline) pricing (the sample and construction of Table 7); policy names as there, plus two additional field-size variants (40, 60) shown only in the left panel. Left: each point is a policy, plotted by predicted returning riders against the mean pre-treatment ability of predicted returners, expressed as seconds per 20-minute lap faster than the average rider. Right: predicted return-rate gain relative to the current structure, by pre-treatment ability decile (1 = fastest, within gender).

9.2 Dynamic microsimulation

The annual exercise holds the pool fixed and so misses the cumulative consequence of retention: a rider kept this season races again next season, accumulates development, and faces another retention draw—and exit is absorbing. I therefore build a dynamic microsimulation that follows the real entering cohorts forward under each design; Appendix I collects

its bridge, decomposition, and validation exhibits. Entry is exogenous (real entrants with their real first-season ability); the endogenous dynamics are division assignment (with the CPT ratchet under the current system), recognition from the pairwise model, persistence calibrated to observed return rates and the causal estimates, and ability evolution. I report the conservative scenario S1 (age effects and mean reversion only) throughout.²

Because the simulation inherits the mechanism question, its conclusions are stated under the count-based mapping and should be read as conditional on that mechanism; under a pure standing mapping the aggregate dynamic gains, like the annual ones, would largely vanish. Two further reconciliations bound the headline. First, the within-simulation *annual* return-rate gap under top-ten recognition is +1.3 to +1.6 percentage points per year—roughly three to five times the annual counterfactual’s +0.3 to +0.4—because the simulation’s synthetic divisions (target size 30) sit in a far higher medal-share regime than the observed fields; the dynamic estimate therefore embeds the simulation’s field structure, not new compounding magic, and inherits the dilution caveat of Section 7.2 with full force. Pricing that caveat rather than gesturing at it: re-running the simulation with the count response replaced by the estimated share-dependent (dilution) schedule, floored at zero, erases the gain entirely (−0.2 percentage points; Appendix Table A17). The simulated medal shares (0.20 under the current design, 0.40 under top-ten recognition, against an observed mean of 0.07) lie far outside the range where the dilution schedule is estimated, so this is a cautionary bound built on an out-of-support extrapolation—but it is the paper’s own schedule, and it says the dynamic gain survives only to the extent recognition holds its value in denser podium regimes. Second, the entrant race-one design of Section 7 finds no first-year response. Re-running the simulation with the entrant response set to zero and the returning-rider response at the strict estimate (+2.3 per finish, applied as 0.023) raises the senior-rate gain to +1.9

²Two errors in an earlier version of the simulation are corrected here. First, the category-placement ratchet was applied asymmetrically under recognition-depth policies, mechanically inflating their effects. Second, the persistence responses to recognition and standing were each calibrated from single-treatment regressions and then applied jointly, double-counting their common variation; the simulation now uses jointly estimated (horse-race) response coefficients. Both corrections reduce the policy effects relative to earlier drafts.

percentage points and the varsity-production gain to +9 percent—above the baseline configuration, which spreads a smaller full-sample response across both populations. The defensible statement is therefore a range: top-ten recognition raises the senior-reach rate by between roughly zero and two percentage points—slightly negative (-0.2) if the dilution schedule is taken at face value in the simulation’s high-share regime, +1.0 under the baseline response, +1.9 if entrants do not respond at all while returning riders respond at the strict estimate. With that conditionality explicit: under the baseline response, the share of entrants who reach their senior year rises from 25.0 to 26.0 percent (+1.0 percentage points), and the production of varsity-level seniors rises by about 4 percent. Decomposing the varsity-level gain (Appendix Figure A12a), nearly all of it flows through the participation channel—more riders surviving to be varsity-eligible—and the small remainder through ability. Consistent with the tier-split reduced form (Appendix Table A3), the extra varsity seniors come disproportionately from middle-ability entrants (Appendix Figure A12c); panel b shows the option value that makes their retention worth having—even mid-decile entrants who reach their senior year clear the varsity bar at 40–50 percent rates. The design lever broadens the top track’s membership; it does not manufacture top-decile ability, and I do not claim otherwise.

Bridging the annual and dynamic magnitudes. Even a modest dynamic gain needs reconciling with the smaller annual one. Appendix Table A17 walks the gap in steps. Level L0 applies the annual gain analytically—compounding the static return-rate change over a career with no field feedback—and yields +0.2 points. Level L1 turns on survival compounding within the simulation’s endogenous fields, holding every rider’s ability frozen: +0.4 points. Level L3 adds ability evolution (full S1): +1.0 points. At the corrected price the amplification is modest and split between the two dynamic channels: retained riders re-enter next season’s fields, where the policy again raises their recognition and survival (L0 to L1), and riders who stay in the sport accumulate ability development that the static calculation ignores (L1 to L3). Propagating the uncertainty in the causal parameter through the simulation (draws

of the persistence response) puts the senior-rate gain between -0.2 and $+0.8$ percentage points—a band whose upper edge sits just below the bridge configuration’s $+1.0$ and whose lower edge includes zero. The dynamic exercise bounds rather than advertises.

Validation and limitations. The simulation is calibrated to reproduce observed return rates by grade, gender, and ability, and validated against the observed senior-participation profile (Appendix Figure A13). Two limitations are stated plainly. First, the simulation under-predicts senior participation—most severely among the fastest deciles, so it also under-captures the ability gradient—because it omits persistent unobserved commitment heterogeneity; the policy conclusions are relative comparisons across designs and are independently corroborated by the reduced forms of Section 5 and the model-light annual counterfactuals, but the levels should not be quoted as forecasts. Second, entry is held fixed: recognition-rich designs plausibly also *attract* entrants, a margin the simulation excludes, making the pipeline gains conservative on that dimension—while the dilution evidence cuts the other way, as in the annual exercise.

10 Conclusion

Ability grouping is usually evaluated as a pedagogy. This paper evaluates it as a retention technology. In a setting where the instruction channel of tracking is absent by construction and the participation margin is a real annual choice, the local competitive success generated by group assignment raises returning riders’ continued participation by about two percentage points per top-five finish: $+1.9$ in the primary simulation design once its residual pools are strictly predetermined, $+2.3$ in the stricter returner-only variant, $+1.4$ in the cleanest-exclusion external design, with the cascade design bounding effects within $[-16, +4]$. The agreement across designs is the finding; the correction that produced it—removing look-ahead that had inflated the estimate roughly threefold—is reported in full rather than absorbed silently, and first-year riders, estimated separately on a strictly pre-

determined design, show no detectable response. Because early performance only weakly predicts who eventually populates the top track—about one-third of varsity seniors were not early standouts—retaining marginal participants has option value, and the reduced forms show early rank luck moves ordinary entrants onto the varsity track without manufacturing elite ability. At the corrected price, design levers that broaden recognition (more and smaller groups, deeper honors) raise annual participation by fractions of a percentage point and compound over careers into single-percentage-point gains in the supply of riders who reach their senior year—real, but modest, and bounded below by approximately zero once the paper’s own recognition-dilution estimate is extrapolated into the denser podium regimes those designs create; under a pure standing mechanism those aggregate gains vanish and only the distributional effects of regrouping remain. The mechanism evidence does not settle what carries the effect: the full-sample horse race favors continuous standing while the strictly predetermined returner design favors counts, the single-race medal moment is a precise null, and recognition dilutes with depth. I have deliberately left the paper agnostic, reporting every policy under both mappings rather than letting a preferred story pick the table.

Nor is this a story about one state. Across six leagues processed through one pipeline, the per-finish price is not statistically distinguishable across leagues—precisely measured in Utah, underpowered elsewhere—and what differs is structure: leagues that subdivide their fields mint roughly twice the recognition that their own merged fields would, a partial-equilibrium accounting worth roughly 0.3 to 1.2 points of aggregate annual persistence at the estimated price, while Colorado, which runs single starts, forgoes that margin entirely. Division structure, in other words, is a recognition-supply policy, and leagues already set it differently.

For education policy, the analogy is direct but must be labeled what it is: speculative extrapolation. The levers studied here—how many ability groups to form, how large to make them, how deep recognition should reach—map onto honor rolls and dean’s lists (how deep should named recognition go?), course leveling (how many tracks, with rank visible

within each?), and gifted-program size (a small elite pond versus broader designation). The findings suggest the participation-relevant content of such structures is the recognition and local standing they generate rather than the labels they confer, that recognition depth is a real but dilutable resource, and that in elective activities—where students can simply not come back—the option value of keeping middling performers engaged is large because early signals are poor. Whether those lessons transfer to settings where stakes, durations, and selection differ is exactly the kind of question the cross-context comparison of Appendix J shows cannot be settled casually.

What remains unresolved: the simulation-versus-external gap among first-year riders; the reported standard errors exclude generated-regressor uncertainty pending a full-pipeline bootstrap; the dynamic projections are conditional on the count mechanism and on calibration choices that richer commitment data could sharpen; and the untracked-benchmark comparison did not deliver the compression pattern a simple tracking story predicts. None of these caveats threatens the core finding—that locally generated competitive success, quasi-randomly assigned, keeps young athletes in the pipeline—but they bound how far the design arithmetic built on it should be trusted.

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Appendix A Rider Identification and Linkage Audit

Procedure. Rider identifiers are constructed in stages. Names are cleaned and standardized; each rider’s grade is inferred from bracket names (e.g., seventh- versus eighth-grade middle-school brackets, Freshman = grade 9) and team fields; an expected graduation year is computed from the grade signals; and same-name riders are split when their records imply conflicting graduation years, different schools, or impossible career-timeline gaps. The procedure resolves 17,729 riders from 148,876 race-result rows. Grade-12 rider-seasons are excluded from all persistence regressions because seniors mechanically cannot return.

Audit. Because a missed link is observationally a dropout, I audit the linkage directly (Table A1). The resolved identifiers contain zero graduation-year conflicts; 2.1 percent of riders are associated with more than two schools (plausible transfers). For each apparent dropout (a non-senior rider who disappears), I search subsequent seasons for near-miss records—same school and gender with a similar name—that could be the same rider under a failed match; 1.2 percent of apparent dropouts have such a candidate, an upper bound on false dropouts from linkage failure. The headline return rate moves from 71.9 percent under loose (name-and-gender) matching to 71.7 percent under the baseline procedure—a 0.2-point difference—and to 64.2 percent under strict (name-gender-school) matching, which mechanically breaks school-switchers’ links. Finally, linkage difficulty is balanced with respect to the identifying variation: an indicator for having a common surname (2.8 percent of rider-seasons) is unrelated to the simulation instrument ($t = 0.1$), so match quality does not covary with the variation that identifies the causal estimates. Out-of-state moves, injuries, and switches to other sports remain observationally equivalent to quitting.

Table A1: Linkage audit

Metric	Value
Riders resolved	17,729
Share of rider IDs with conflicting graduation years	0.0%
Share of rider IDs with more than two schools	2.1%
Apparent dropouts (non-senior)	7,664
Near-miss candidate rate among dropouts (upper bound on false dropouts)	1.2%
Return rate, loose matching (name + gender)	71.9%
Return rate, baseline (v2 identifiers)	71.7%
Return rate, strict matching (name + gender + school)	64.2%
Common-surname share of rider-seasons	2.8%
Common-surname balance on simulation instrument (t)	0.1

Notes: Strict matching mechanically codes school-switchers as dropouts, so its lower return rate is expected. The loose-versus-baseline gap of 0.2 pp bounds the sensitivity of the outcome to over-splitting.

Appendix B Pre-Treatment Skill

Skill is the mean residual from a regression of log average lap time on venue, year, category, and laps ($R^2 = 0.41$), measured using only pre-treatment (prior-season) information and shrunk toward grade $\times$ gender $\times$ year cell means by empirical Bayes. The measure is highly stable (season-to-season autocorrelation 0.82). The underlying season-mean residual is itself reliable rather than noise-driven: its split-half correlation across a rider’s odd- and even-numbered races within a season is 0.74, and its year-over-year correlation among returning riders is 0.60.

Entrant timing. First-season entrants have no prior season, so an entrant ability measure built from the full first season would embed the same races that generate first-season recognition—post-treatment variation. All long-run analyses therefore use a race-one-only construction for entrants: ability and expected recognition are computed from the first race alone, with outcomes defined over subsequent races. Table A2 compares the long-run reduced forms across constructions. The varsity-margin results are robust—+32.0 per unit of race-one top-five rate ($t = 7.2$), +14.8 per race-one expected count ($t = 8.6$), +7.1 in the

original (full-first-season) variant ($t = 7.1$)—while the effect on cumulative seasons raced is a null under every timing construction, and I treat it as such in the text.

Table A2: Long-run reduced forms under alternative entrant-timing constructions

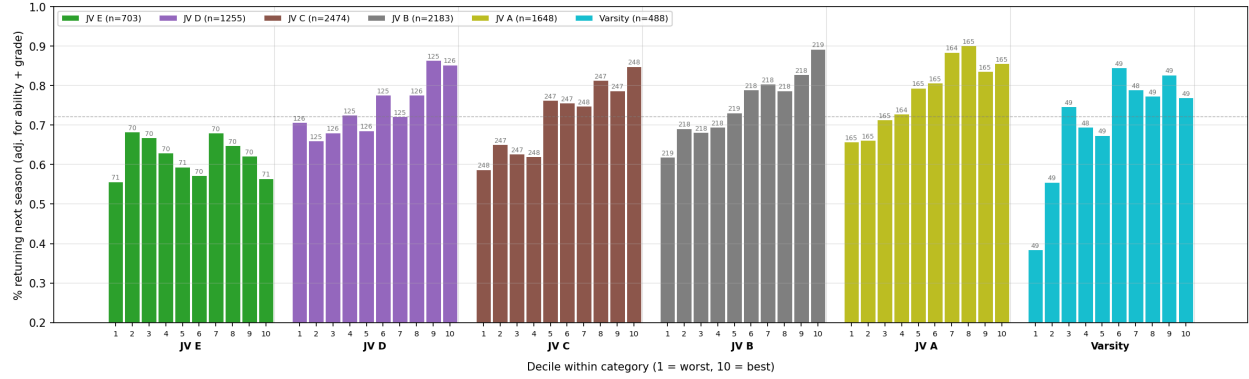
Outcome	Race-1 rate		Race-1 count		Original (full season)	
	Coef.	t	Coef.	t	Coef.	t
Future seasons (count)	+0.06	0.5	−0.01	−0.3	+0.01	0.5
Ever senior (pp)	+7.3	1.6	+3.1	1.7	+1.2	1.2
Ever varsity (pp)	+32.0	7.2	+14.8	8.6	+7.1	7.1
Becomes top talent (pp)	+30.7	9.0	+12.2	8.6	+4.9	8.5
Varsity/elite as senior (pp)	+21.4	6.2	+10.1	7.1	+4.4	6.4

Notes: Specifications as in Table 2. $N = 8,458$ for the race-one variants, 9,249 for the original variant. Coefficient units differ across variants (per unit rate, per expected count); signs and significance are the comparable objects.

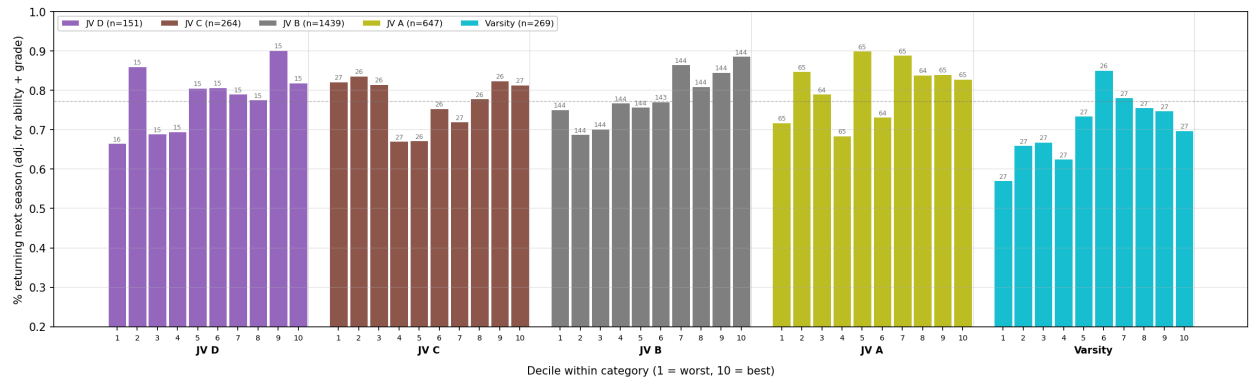
Split-sample construction. Because ability is an estimated, shrunken regressor entering both the control and the instruments, Appendix Table A13 includes a split-sample specification in which ability is estimated from odd-numbered races and the instrument is built on even-numbered races, eliminating correlated estimation error by construction; the estimate is +4.0 ($t = 6.3$), attenuated as expected under classical measurement error. Results are also robust to using unshrunk ability.

Appendix C Descriptive Detail

This appendix collects the descriptive exhibits summarized in Section 5: the Utah within-division return gradient (ability- and grade-adjusted), the winner’s boost at fixed ability, the early-performance predictors of senior varsity participation, and the tier-split race-one reduced forms.



(a) Boys



(b) Girls

Figure A1: Return rate by within-division performance decile (Utah)

Notes: Share returning the next season (adjusted for pre-treatment ability and grade) by finishing decile within division (1 = worst, 10 = best), separately by division. The dashed line is the overall adjusted mean. The upward within-division gradient—present for both genders and net of ability—is the descriptive analogue of the paper’s causal result.

Table A3: Race-one rank luck by initial-ability tier: effects per SD of the instrument

Initial tier	Effect per SD of instrument (pp)			Instr. mean (SD)	Base varsity (%)	N
	Ever senior	Ever varsity	Top decile			
Elite (top 20%)	+2.6 (2.1)	+9.2 (7.7)	+7.2 (8.8)	0.60 (0.54)	41.9	1,8
Middle (deciles 3–6)	−0.4 (−0.5)	+2.4 (3.0)	+0.3 (1.7)	0.21 (0.30)	23.9	4,4
Bottom 40%	+0.8 (0.7)	+2.1 (1.9)	+0.0 (1.0)	0.02 (0.08)	24.3	2,1

Notes: Each cell reports the effect of a one-SD (within tier) increase in race-one expected top-five count on the column outcome in percentage points; *t*-statistics in parentheses (robust SEs). Conditioning as in Table 2. “Base varsity” is the tier’s mean ever-racing-varsity. Winsorizing the instrument at its 99th percentile leaves all conclusions unchanged. The top-decile column elite-tier effect partly reflects mechanical persistence of high initial ability; the relevant contrast is its absence in the other tiers.

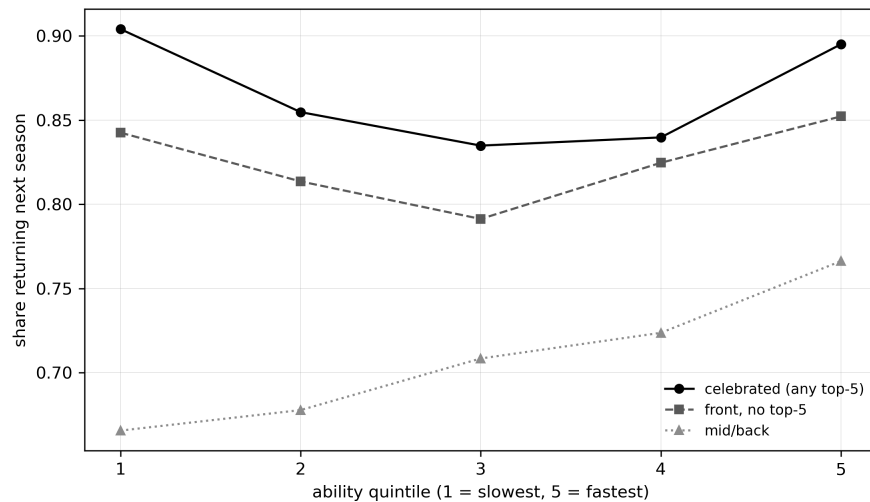


Figure A2: The winner’s boost at fixed ability

Notes: Share returning the next season by global ability quintile (within season×category, pooled across divisions) and local experience in the rider’s own division. “Celebrated” = at least one top-five finish during the season; “front, no top-5” = top three standing deciles within the division without a top-five; “mid/back” = all others. Utah divided categories, persistence sample. Cells with fewer than 30 rider-seasons suppressed.

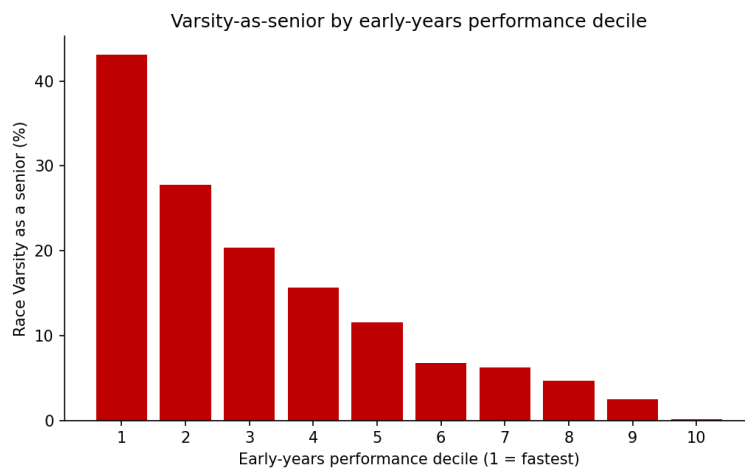
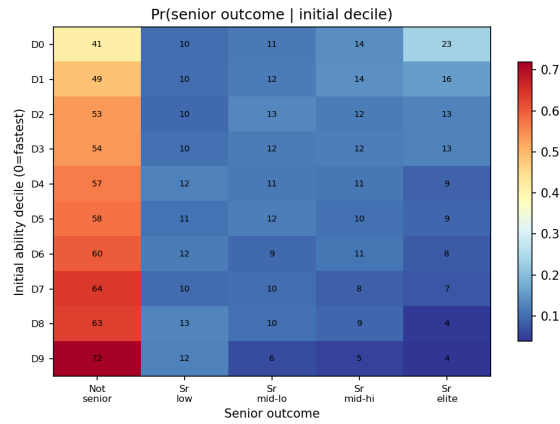
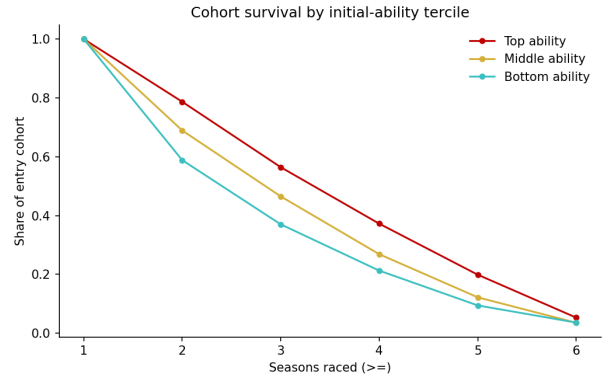


Figure A3: Racing varsity as a senior, by early-years performance decile

Notes: Riders first seen in grades 7–9 with an observable senior year. About one-third of varsity seniors come from outside the top three early-performance deciles; 43 percent of the fastest early decile race varsity as seniors.



(a) Initial decile $\rightarrow$ senior outcome



(b) Survival by initial-ability tercile

Figure A4: Early ability under-predicts eventual success.

Notes: Cohorts with an observable senior year (graduation ≤ 2025). Initial ability is the first-season pre-treatment skill decile within gender.

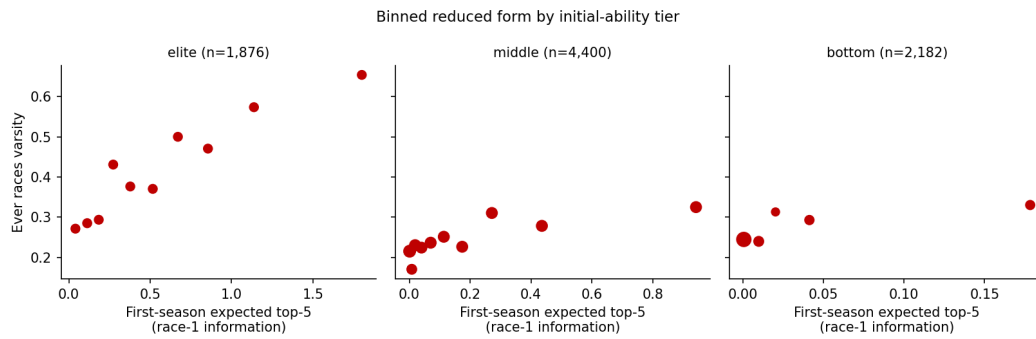


Figure A5: Binned reduced forms by initial-ability tier

Notes: Binned means of long-run outcomes against race-one expected recognition, by initial-ability tier, corresponding to Table A3. The plots show the effects are not driven by thin right-tail leverage within tiers.

Appendix D The Expected-Rank Engine

The pairwise model treats course-adjusted race time as ability plus a mean-zero shock with SD σ ; rider j beats i with probability $\Phi((A_i - A_j)/(\sigma\sqrt{2}))$. Expected rank and fraction-beaten are computed in closed form; win and top- k probabilities are computed by Monte-Carlo ranking so that exactly k riders are top- k in each draw. The Monte-Carlo draws resample *empirical* race-day residuals rather than normal deviates, so the fat tail of race-day disasters (mechanicals, crashes) is reflected in the top- k tail probabilities by construction. Figure A6 reports calibration: Spearman correlation of predicted and realized rank 0.74; top-five AUC 0.87.

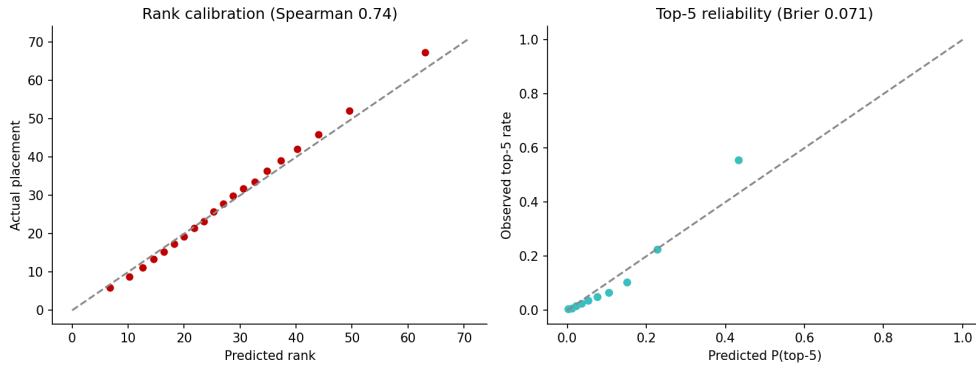


Figure A6: Predicted vs. actual rank and top-five reliability

Notes: Out-of-sample calibration of the pairwise engine: predicted versus realized finishing rank (Spearman 0.74) and top-five reliability (AUC 0.87). Monte-Carlo draws resample empirical race-day residuals.

Appendix E The Cascade Design: Density, Inference, and Compilers

Figure A7 shows the density of the CPT running variable; the absence of bunching at division cutoffs is consistent with the identifying assumption (riders and coaches do not precisely manipulate CPT rank around cutoffs). Table A4 collects the weak-instrument-robust inference: the 95 percent Anderson–Rubin set for the top-five count effect is $[-16, +4]$

percentage points at baseline and $[-32, +9]$ in a donut specification excluding riders within two positions of a boundary; randomization inference re-drawing cutoff placements 200 times yields $p = 0.46$ for the reduced form (-1.3 pp). A stacked-boundary design (riders within 2–8 CPT positions of a cutoff, boundary fixed effects, distance trends) finds no first stage at the predicted discontinuities ($t \in [-0.7, +0.5]$), consistent with the rule reproducing only 42 percent of assignments near cutoffs. The design’s correlation with predetermined commitment is small (prior racing: -0.010 , $t = -2.2$; Table A10).

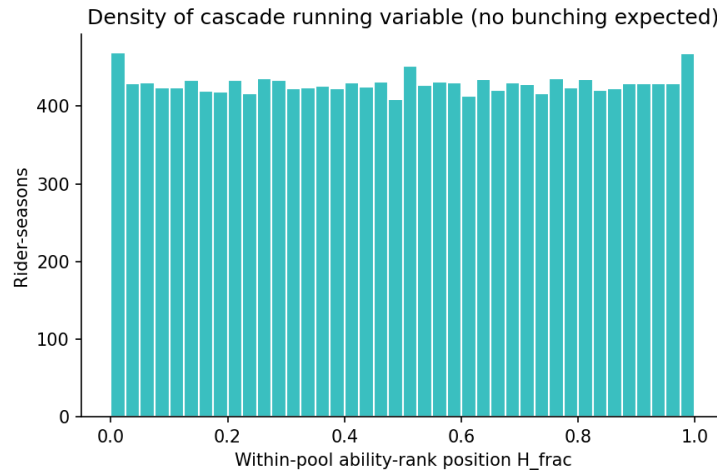


Figure A7: Density of the cascade running variable

Notes: Density of the CPT running variable; the absence of bunching at division cutoffs is consistent with no precise manipulation of placement rank.

Table A4: Cascade inference

Specification	AR 95% set (pp)	N
Baseline	$[-16.0, +4.0]$	17,164
Donut (≥ 2 positions from boundary)	$[-32.0, +9.0]$	16,126
Randomization inference (200 permutations of cutoffs)	RF = -1.3 pp, $p = 0.46$	

Notes: Anderson–Rubin sets for the effect of one top-five finish on next-season participation under the cascade design.

Table A5 reports the slot-ladder promotion design built from the placement table’s own logic (Section 6): returning riders default to their period- T division and vacated seats are filled in period- T points order, giving each returner below the top division a predicted promo-

tion cutoff. The first stage is stable across bandwidths; the persistence effect at the margin is a bounded zero for the power reasons discussed in the text.

Table A5: The slot-ladder promotion design

	± 5 ranks	± 8 ranks	± 12 ranks	all (quadratic)
First stage: actually promoted	+0.12 ($t = 2.2$)	+0.11 ($t = 2.4$)	+0.14 ($t = 2.6$)	+0.07 ($t = 2.2$)
$T+1$ local standing (frac. beaten)	-0.03 ($t = -1.0$)	-0.03 ($t = -1.1$)	-0.04 ($t = -1.4$)	+0.02 ($t = 0.6$)
Persistence (raced $T+2$, pp)	-7.2 (11.4) ^a	+2.3 (8.2)	-1.4 (10.0)	+5.0 (4.0)
N (riders near cutoff)	924	1,428	2,030	3,669
Validation: ratchet violations 2.3%; promotion-points-rank Spearman -0.64 , right-signed in every cell; capacity absorption -0.46 (vs. $1.0 =$ full absorption); 96 cutoff cells.				

Notes: Cell (pool $\times$ year $\times$ source-division) fixed effects with side-specific linear distance trends (± 12 : quadratic); standard errors clustered by pool-year. Persistence rows use the non-senior, non-censored subsample (smaller N).

^a Persistence point estimates are noisy across bandwidths; the design is power-limited, not informative, at this margin (see text).

Table A6: Supporting estimates referenced in the text

	Coef.	(s.e.)	t	N
<i>Panel A. Graduation-slack reduced form (exposed riders, margins B+C stacked)</i>				
Senior share above $\rightarrow$ raced next season	+0.87	(0.32)	2.7	10,553
balance: exposed riders' ability	-0.02	—	-0.2	11,048
balance: share first-year	+1.18	—	1.8	11,048
<i>Panel B. Recognition dilution (2SLS, persistence on top-five count)</i>				
Top-five count	+0.016	(0.006)	2.6	28,434
Top-five $\times$ medal share (demeaned)	-0.150	(0.046)	-3.3	28,434
<i>Panel C. Cascade-eligible subsample (returning riders, division brackets)</i>				
Simulation design: top-five count	+0.022	(0.009)	2.4	14,958
External design: top-five count	+0.009	(0.011)	0.8	14,958

Notes: Panel A: reduced form of next-season participation on the senior share of last year's upper-division riders, for riders not protected at the margin; pool and year-by-margin fixed effects, lag size, pool sizes, ability deciles, and experience controls; clustered by pool-year. Balance rows regress the listed characteristic on the instrument. Panel B: 2SLS as in Table 3 with the medal-share interaction instrumented by the instrument-share interaction. Panel C: the main specifications re-estimated on the cascade-eligible subsample.

Table A7 characterizes compliers via subgroup first stages. The simulation and external instruments move recognition broadly (first stages of 0.7 to 1.3 across subgroups). The cascade's first stage is concentrated among returning riders (0.29) and the fast initial-ability tercile (0.56), and is weak for girls (0.14) and the slow tercile (0.08): its LATE is for fast,

returning riders near division-size cutoffs—the population on which all three designs agree (Section 7). No subgroup shows a sign-reversed first stage for any design, the testable implication of monotonicity.

Table A7: Subgroup first stages (complier characterization)

Subgroup	Simulation		External		Cascade	
	Coef.	t	Coef.	t	Coef.	t
All	0.90	60.7	1.07	44.3	0.27	7.9
Boys	0.86	49.1	1.08	34.7	0.31	7.8
Girls	0.94	31.7	1.03	24.3	0.14	2.0
First-year	0.91	54.0	1.31	40.6	0.26	3.7
Returning	0.86	31.7	0.89	26.1	0.29	7.5
Fast tercile	0.96	51.5	1.09	34.7	0.56	9.0
Slow tercile	0.73	13.6	0.68	11.6	0.08	2.4

Notes: First-stage coefficients of realized top-five count on each instrument, by subsample, with controls as in Table 3. Cascade subsamples restricted to cascade-eligible rider-seasons.

Appendix F Mechanism Detail

This appendix collects the heterogeneity and mechanism exhibits summarized in Section 7: the gender and experience splits, the pooled fifth/sixth-place regression discontinuity, the nonparametric dose–response and close-finish placebo, the count-to-standing translation, and the channel decomposition.

Table A8: Heterogeneity: gender and experience

Sample	Top-five count (pp per finish)			Fraction beaten (pp, 0→1)	
	Simulation	External	Strict returner	Simulation	N
Boys	+1.7 (0.8) [1.0]	+0.0 (1.0) [1.2]	−1.9 (2.0) [2.1]	+8.9 (1.9) [2.8]	22,097
Girls	+1.5 (1.0) [1.3]	+1.0 (1.3) [1.9]	+0.8 (2.0) [2.0]	+6.5 (3.4) [4.2]	6,337
First-year	+0.5 (0.7) [1.1]	−1.1 (0.9) [1.5]	—	+11.4 (2.2) [3.8]	14,496
Returning	+1.1 (0.9) [1.1]	+2.5 (1.1) [1.2]	+1.2 (1.3) [1.3]	−0.2 (2.7) [2.9]	13,938

Notes: 2SLS estimates of the effect on next-season participation, by subsample. Simulation and external columns as in Table 3 (demeaned pools); the strict returner column uses the per-median-season treatment with $\leq t - 1$ demeaned pools and exists only for returning riders (Boys $N = 10,748$; Girls $N = 3,190$; Returning $N = 13,938$). SEs clustered by rider in parentheses, by field in brackets. All designs are positive for returning riders; the first-year margin is a null in the entrant race-one design of Table 3; see text.

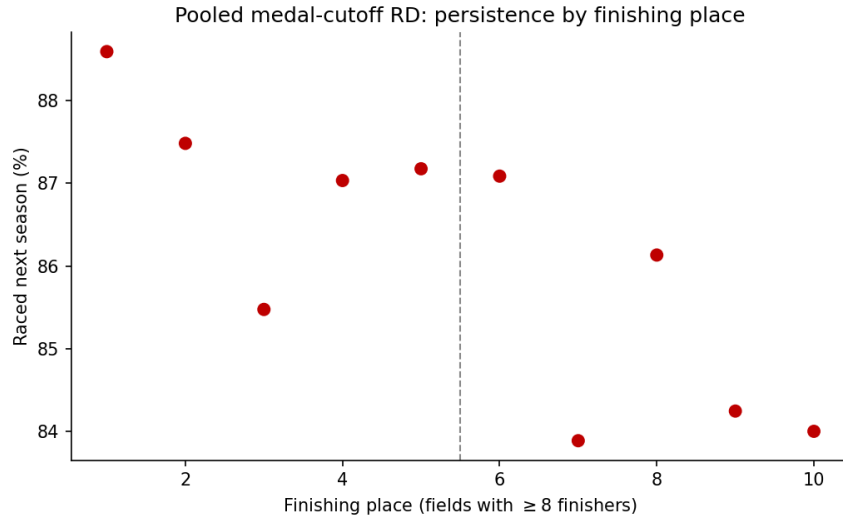
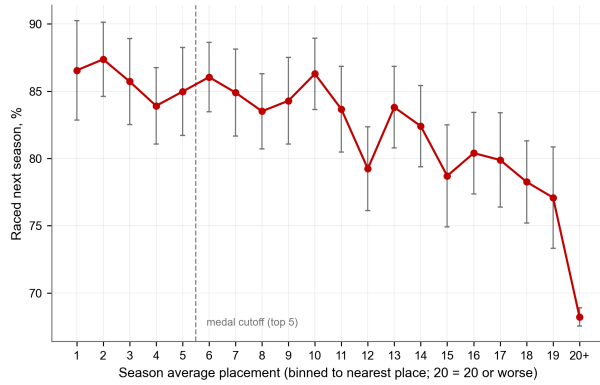
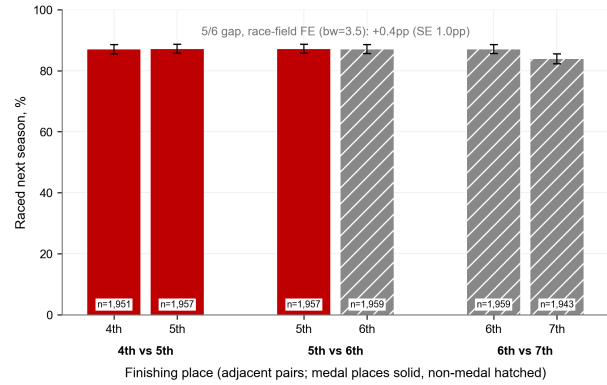


Figure A8: Pooled regression discontinuity at the fifth/sixth-place cutoff

Notes: Next-season participation against single-race finishing position relative to the medal cutoff, pooled across all races and seasons. The discontinuity is +0.4 pp (SE 1.0) at bandwidth 3.5; $MDE_{80} = 2.5$ pp.



(a) Persistence by average placement



(b) Close-finish placebo

Figure A9: The effect is cumulative over the season, not a single-race threshold.

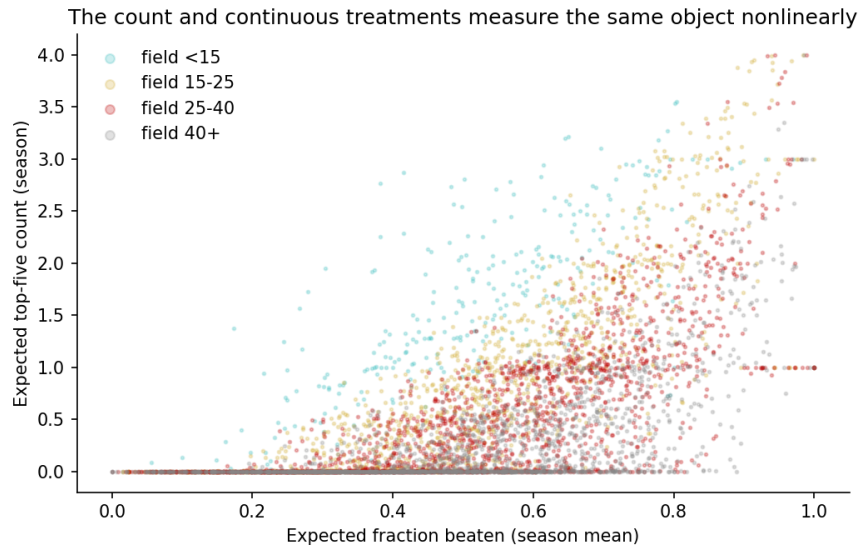


Figure A10: Translating counts into standing through the pairwise model

Notes: Implied 0→1 fraction-beaten effects from the count estimate, by field size, versus the directly estimated standing effect. At typical field sizes a 0→1 standing change corresponds to roughly 1.8 top-five finishes per season.

Table A9: Channel decomposition: label, peers, field size

Specification	Coefficient (pp)	t	N
Baseline, full sample	+1.9	3.3	28,434
Baseline, division brackets only	+2.1	2.3	14,931
+ division-label fixed effects	+4.0	4.1	14,931
+ peer-quality controls	+0.9	0.8	14,931
+ field size	−0.5	−0.5	14,931
Standing treatment, all channel controls	+12.9	3.2	14,931
External instrument: baseline (full sample)	+1.4	2.0	28,434
External instrument: baseline (div. brackets)	+0.9	0.8	14,931
+ division-label fixed effects	+2.6	2.4	14,931
+ peer-quality controls	−0.9	−0.7	14,931
+ field size	−4.2	−3.2	14,931

Notes: 2SLS estimates (simulation design, demeaned pools) of the top-five-count effect on next-season participation (top panel) and, for comparison, the external instrument (bottom panel), progressively absorbing the non-recognition contents of division assignment. Peer-quality controls: division mean and SD of ability, own-school teammate count and mean strength. Rider-clustered SEs. The “standing treatment” row replaces the count treatment with fraction beaten under the full control set. Peer quality and field size are inputs to the expected-recognition instruments, so the later rungs absorb identifying variation along with the channels they name; sign flips there (external −4.2; standing +12.9) mark the point where the decomposition stops being informative rather than evidence of perverse effects.

Appendix G Robustness Appendix

This appendix collects, as numbered tables, every robustness claim made in the text.

G.1 Balance and pre-trends

Table A10 regresses predetermined variables on each instrument, conditional on the baseline controls. The first panel—prior-season participation regressed on the season- s instruments—is the pre-trends placebo: every instrument shows a small negative residual correlation (coefficients -0.008 to -0.021 ; t from -0.9 for the external standing instrument to -7.5 for the simulation count), precisely estimated but an order of magnitude too small, and of the wrong sign, to generate the estimated effects. Lagged performance panels load on all instruments conditional on ability deciles, as functions of lagged ability must—the strict returner instrument still predicts the lagged top-five count ($t = 13.0$), and the joint tests in Panel E reject for every instrument, including the strict and fixed-reference constructions built to be predetermined. Balance on lagged history is not restored, and the paper does not claim it is; the operative checks are the stability of the estimates under much richer $f(A)$ (Table A13) and the direct lagged-history conditioning reported in Section 6, which leaves the strict returner estimate at $+1.0$ (s.e. 1.3). Table A11 isolates the experience exhibit: without experience controls the standing instrument is wildly imbalanced on prior participation (-0.70 , $t = -39.8$); with them the imbalance falls by a factor of nearly fifty (-0.015 , $t = -2.8$).

G.2 Attendance purge

Table A12 reports the headline estimates under the old (raw seasonal count) and new (per-race rate $\times$ median three-race season) treatment definitions, plus a fully rate-on-rate specification.

Table A10: Balance of predetermined variables across instruments (conditional on controls)

Predetermined variable	Instrument	Coef.	t	N
<i>A.</i> Raced previous season	Simulation top-five (std)	−0.015	−7.5	32,131
	Simulation strict (returners)	−0.021	−4.9	17,042
	Simulation fraction-beaten	−0.015	−2.8	32,131
	External top-five (std)	−0.013	−5.0	32,131
	Fixed-ref external (returners)	−0.016	−3.4	17,015
	External fraction-beaten	−0.008	−0.9	32,131
	Cascade top-five	−0.010	−2.2	17,164
<i>B.</i> Lagged top-five count	Simulation top-five (std)	+0.338	12.3	16,301
	Simulation strict (returners)	+0.313	13.0	16,298
	External top-five (std)	+0.451	13.1	16,301
	Fixed-ref external (returners)	+0.350	10.9	16,271
	Cascade top-five	+0.379	9.1	11,809
<i>C.</i> Lagged fraction beaten	Simulation top-five (std)	−0.001	−0.2	16,301
	Simulation strict (returners)	−0.004	−0.9	16,298
	External top-five (std)	+0.012	1.8	16,301
	Fixed-ref external (returners)	+0.002	0.3	16,271
	Cascade top-five	+0.043	4.6	11,809
<i>D.</i> Lagged races entered	Simulation top-five (std)	−0.095	−6.3	16,301
	Simulation strict (returners)	−0.054	−4.0	16,298
	External top-five (std)	−0.146	−8.9	16,301
	Fixed-ref external (returners)	−0.098	−6.0	16,271
	Cascade top-five	−0.063	−2.7	11,809
<i>E.</i> Joint F (five history variables)	Simulation top-five (std)	589	$p < 0.001$	32,131
	Simulation strict (returners)	524	$p < 0.001$	17,042
	Simulation fraction-beaten	5,269	$p < 0.001$	32,131
	External top-five (std)	166	$p < 0.001$	32,131
	Fixed-ref external (returners)	929	$p < 0.001$	17,015
	External fraction-beaten	1,203	$p < 0.001$	32,131
	Cascade top-five	69	$p < 0.001$	17,164

Notes: Panels A–D: each row regresses the predetermined variable on the instrument, conditional on the baseline control set (ability deciles, experience controls, region×year FE). Panel A (prior-season participation on current instruments) is the pre-trends placebo. Lagged-performance correlations are expected: the instruments are functions of lagged ability; see text. Panel E reports the joint F -statistic from regressing each instrument on the five predetermined history variables jointly—rejection everywhere is the honest summary, and the paper’s response is $f(A)$ enrichment (Table A13) plus direct lagged-history conditioning (Section 6), not a claim of balance. Rider-clustered t -statistics.

Table A11: The experience confounder: balance of prior participation on the standing instrument

Controls	Coef.	<i>t</i>	<i>N</i>
Skill and year fixed effects only	−0.699	−39.8	32,131
+ experience controls (races entered, first-year status)	−0.015	−2.8	32,131

Notes: Prior-season participation regressed on the simulation fraction-beaten instrument. A fast newcomer mechanically dominates an easy field, so expected standing is correlated with being new—which independently predicts attrition—until experience is controlled.

Table A12: Purging attendance from the count treatment

Specification	2SLS	(rider SE) [field SE]	First-stage <i>F</i>	<i>N</i>
Simulation: raw count (old)	+0.8	(0.5) [0.7]	5,669	28,434
Simulation: standardized (new)	+1.9	(0.6) [0.7]	3,985	28,434
Simulation: rate-on-rate	+5.6	(1.7) [2.2]	6,566	28,434
External: raw count (old)	+1.6	(0.6) [0.9]	2,474	28,434
External: standardized (new)	+1.4	(0.7) [1.0]	1,987	28,434
External: rate-on-rate	+4.5	(2.2) [3.1]	2,405	28,434

Notes: Outcome is next-season participation, pp. “Raw count” sums top-five finishes over attended races (attendance embedded); “standardized” is the per-race rate scaled to a median three-race season; “rate-on-rate” instruments the per-race top-five rate with the expected per-race rate (coefficient per 0→1 rate change). Races entered controlled in all rows. Controls as in Table 3.

G.3 Ability-control and fixed-effect ladders

Tables A13 and A14 report the estimate under progressively richer ability controls and fixed effects.

Table A13: Robustness to the ability control $f(A)$

Specification	2SLS	(rider SE) [field SE]	First-stage F	N
Simulation: ability-decile FE (base)	+1.9	(0.6) [0.7]	3,985	28,434
Simulation: 20 percentile bins	+1.8	(0.6) [0.8]	3,883	28,434
Simulation: $A \times \text{grade} \times \text{gender}$	+2.0	(0.6) [0.8]	3,770	28,434
Simulation: split-sample (odd/even races)	+4.0	(0.6) [0.8]	2,183	28,434
External: ability-decile FE (base)	+1.4	(0.7) [1.0]	1,987	28,434
External: 20 percentile bins	+1.1	(0.7) [1.0]	1,901	28,434
External: $A \times \text{grade} \times \text{gender}$	+1.4	(0.7) [1.0]	1,897	28,434
Strict returner: 20 percentile bins	+1.3	(1.3) [1.4]	760	13,938
Strict returner: $A \times \text{grade} \times \text{gender}$	-2.5	(1.5) [1.6]	648	13,938

Notes: Top-five count effect on next-season participation, pp. Simulation rows use the demeaned, look-ahead-free pools. The split-sample row estimates ability from odd-numbered races and builds the instrument on even-numbered races, eliminating correlated estimation error; attenuation is expected under classical measurement error. The strict returner rows use the $\leq t - 1$ demeaned construction (per-median-season treatment) on the returner sample; the sign flip under the full interaction battery is the fragility reported in Section 6.

Table A14: Robustness to fixed effects

Specification	2SLS	(rider SE) [field SE]	First-stage F	N
Simulation: year FE only	+1.8	(0.6) [0.8]	4,002	28,434
Simulation: region $\times$ year (base)	+1.9	(0.6) [0.7]	3,985	28,434
Simulation: school $\times$ year (absorbed)	+1.6	(0.6) [0.8]	3,585	27,900
External: year FE only	+1.5	(0.7) [1.0]	1,997	28,434
External: region $\times$ year (base)	+1.4	(0.7) [1.0]	1,987	28,434
External: school $\times$ year (absorbed)	+1.4	(0.8) [1.0]	1,880	27,900

Notes: Top-five count effect on next-season participation, pp. The school $\times$ year specification compares riders on the same team in the same season. Singleton school-years are dropped ($N = 27,900$).

G.4 Speed outcomes: no performance response

Table A15 supports the participation-not-performance discussion in Section 7.2.

Table A15: Recognition does not move measured speed

Specification	Coef.	(s.e.)	First-stage F	N
Next-season pace, baseline ability controls	+1.58	(0.29)	599	9,860
Next-season pace, $A \times \text{grade} \times \text{gender}$	+0.06	(0.34)	440	9,860
In-contention race pace (rider $\times$ season FE)	-0.93	(0.42)	—	65,647

Notes: Units: percent of lap speed (course-adjusted log average lap time $\times (-100)$; venue, season-year, and gender $\times$ grade benchmarks; penalty rows corrected to riding time). Rows 1-2: 2SLS effect of an additional top-five finish on next-season pace, external-opponent instrument, returning riders with at least two next-season races; the baseline-control row documents the ability-control artifact that the interacted controls eliminate. Row 3: race-level within-rider $\times$ season estimate of riding faster when the day's start field puts the rider in top-five contention (expected rank ≤ 7 from prior-season skill); the full expected-rank profile is flat within ± 0.15 percent. The nominally negative row-3 estimate is the coarse single-indicator version; the flat bin-by-bin profile is the informative summary, and no bin shows the positive effort response an incentive story predicts. Rider-clustered SEs. The full design space (alternative pace measures, censoring, selection bounds, promotion-cutoff RD) is archived with the replication materials.

Appendix H Cross-League Detail

This appendix collects the supply-accounting exhibits summarized in Section 8.

Table A16: Division structure as recognition supply

League	Top-five finishes per 100 rider-seasons		Structure multiplier	Implied return-rate contribution (pp)
	Actual (with divisions)	Merged (one start per race)		
Utah	36	15	2.35×	0.6–1.1
Colorado	32	31	1.04×	0.0–0.1
Minnesota	34	16	2.16×	0.5–1.0
Wisconsin	42	22	1.90×	0.5–1.1
Virginia	57	44	1.29×	0.3–0.7
Alabama	53	30	1.75×	0.6–1.2

Notes: Partial-equilibrium accounting, not a behavioral prediction. The counterfactual merges each race's category-gender field into one start, attendance held fixed; championship races excluded; rates computed over each league's full panel of ranked, non-championship starts. The merged column awards exactly five slots per merged start. The implied contribution prices each league's difference in slots at the pooled per-finish range across the two pooled IV designs (simulation +2.7, external-opponent +5.4 pp; both shown in Table 6) and assumes the price is invariant to the merge; the within-paper dilution estimate implies this overstates the structural gain (see text), and non-Utah leagues are priced at a pooled estimate that Utah largely carries.

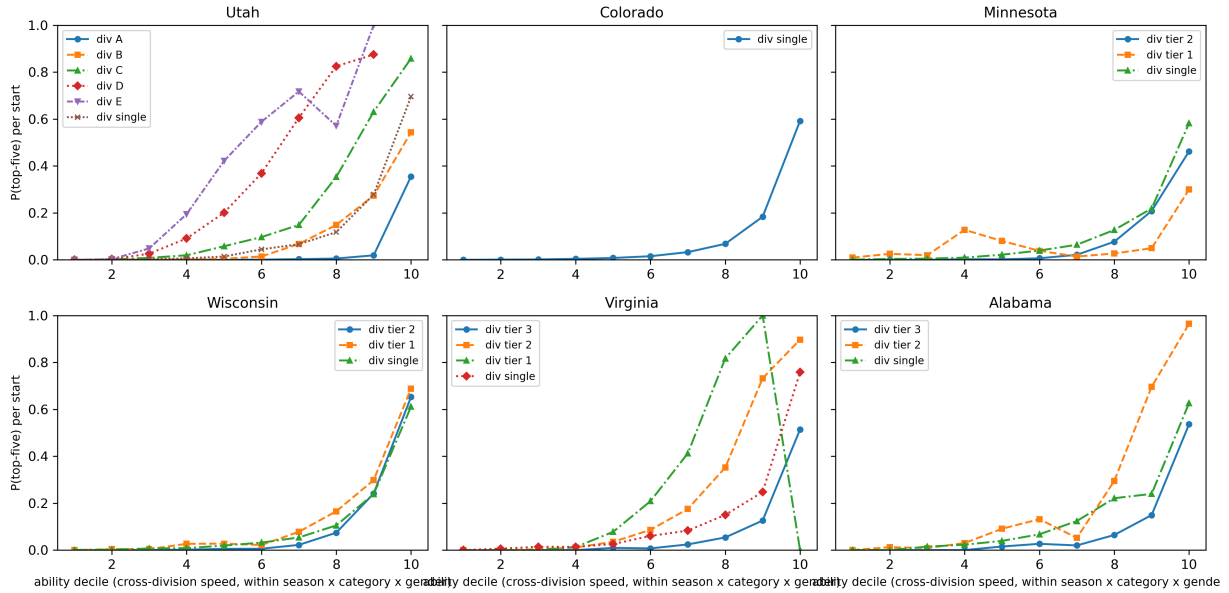


Figure A11: The incidence of recognition supply: top-five probability by ability and division

Notes: Probability of a top-five finish per start by ability decile (cross-division speed within season×category×gender) and division, per league. Where divisions exist, the curves separate: riders of identical measured ability face sharply different odds depending on their division, and mid-ability riders in lower divisions reach probabilities that mid-ability riders in merged or single-start fields (Colorado) reach in under 1 percent of starts.

Appendix I Simulation Detail

This appendix collects the bridge, decomposition, and validation exhibits for the dynamic microsimulation of Section 9.2.

Table A17: Bridging the annual estimate to the dynamic simulation (top-10 recognition, S1)

Level	Senior-reach rate (%)		Gain (pp)
	Current	Top-10 recognition	
L0: Analytic compounding of the annual gain	17.9	18.0	+0.2
L1: Survival compounding in simulated fields (abilities frozen)	23.7	24.2	+0.4
L3: Full simulation, S1 ability evolution	25.0	26.0	+1.0
L3, dilution-priced count response	25.0	24.8	−0.2
L3, entrant response zero; returning at strict (0.023)	25.3	27.3	+1.9

Notes: L0 compounds the annual (cross-sectional) return-rate gain over a career with no feedback. L1 simulates careers with retention feeding back into field composition and recognition, with abilities frozen at entry values. L3 is the full S1 simulation; levels are numbered as in the replication code, where L2 (ability evolution) and L3 (the full protocol) coincide. Varsity-level production rises 0.9 percent at L1 and 4.2 percent at L3, so at the corrected price the amplification is split between survival compounding and ability evolution rather than driven by either alone. The dilution-priced row replaces the count response with the share-dependent schedule of Section 7.2 (floored at zero, count-only, centered at the observed mean medal share); its levels differ from L3 because the response function differs everywhere, and the simulated medal shares (0.20 current, 0.40 top-ten, versus 0.07 observed) lie outside the schedule’s estimated support; varsity-level production falls 1.6 percent. The entrant-zero row sets the first-year response to zero (the entrant design’s null) and the returning-rider response to the strict estimate (0.023 per finish); varsity-level production rises 8.8 percent. Range across causal-parameter draws: −0.2 to +0.8 pp.

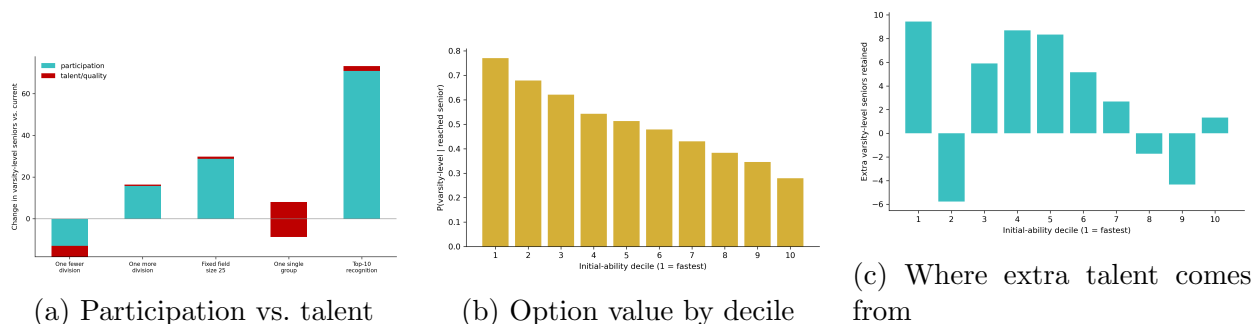


Figure A12: The varsity-level talent gain from broader recognition is a participation effect concentrated among middle-ability entrants.

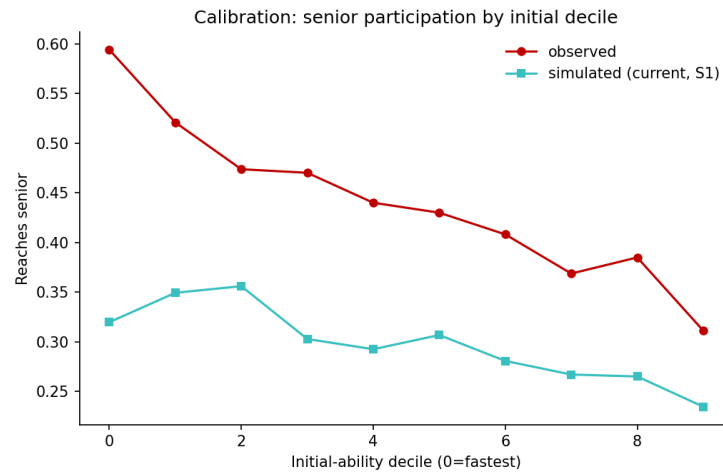


Figure A13: Simulation validation: senior participation by initial decile

Notes: Observed vs. simulated (current policy, scenario S1) senior participation. The simulation underpredicts senior participation, most severely among the fastest deciles, so the policy conclusions are relative comparisons across designs rather than level forecasts.

Appendix J The Untracked Benchmark: A Descriptive Comparison to Alpine Racing

If ability grouping protects low-ranked participants from dropping out, a parallel sport *without* within-age ability tiers should display steeper persistence-by-rank gradients. Utah offers a natural candidate: Intermountain Division alpine ski racing, which serves the same ages in the same state but groups athletes only by age and gender, with no ability divisions. I construct harmonized return-rate gradients—same outcome (raced the next season), same decile construction (within age-group $\times$ gender, pooled across birth years), both genders, raw and ability-adjusted—for alpine and for two MTB analogues: the within-division gradient (the tracked object) and the within-grade-gender gradient pooled across divisions (the gradient MTB would display if it were untracked).

The comparison, shown in Figure A14 and Table A18, does *not* deliver the clean contrast the tracking framing predicts, and I report it as descriptive context rather than evidence. The raw alpine gradients are large: a 24.9-percentage-point return-rate gap between top and bottom decile for boys and 20.0 for girls. But the harmonized MTB analogues are similar or larger (raw pooled: 32.5 boys, 20.1 girls; raw within-division: 29.6 and 20.9), and after ability adjustment the alpine gradients (14.7 and 5.6) are *smaller* than the MTB ones (within-division adjusted: 27.8 and 17.8). Taken at face value, riders’ persistence in this league is at least as rank-sensitive within their ability-matched divisions as alpine racers’ persistence is within untracked age groups—the opposite of a simple compression story. The honest reading is that cross-sport comparisons of this kind are confounded by everything that differs between the sports: cost structure (alpine is dramatically more expensive, which may itself flatten rank-sensitivity by selecting committed families), club versus school institutions, season structure, and the underlying populations. The within-league causal evidence and the same-sport cross-league comparison of Section 8, not this comparison, carry the paper’s claims; the alpine benchmark is included because the tracking framing predicts a pattern

here, a referee would ask for it, and reporting that it does not cleanly appear is more informative than omitting it.

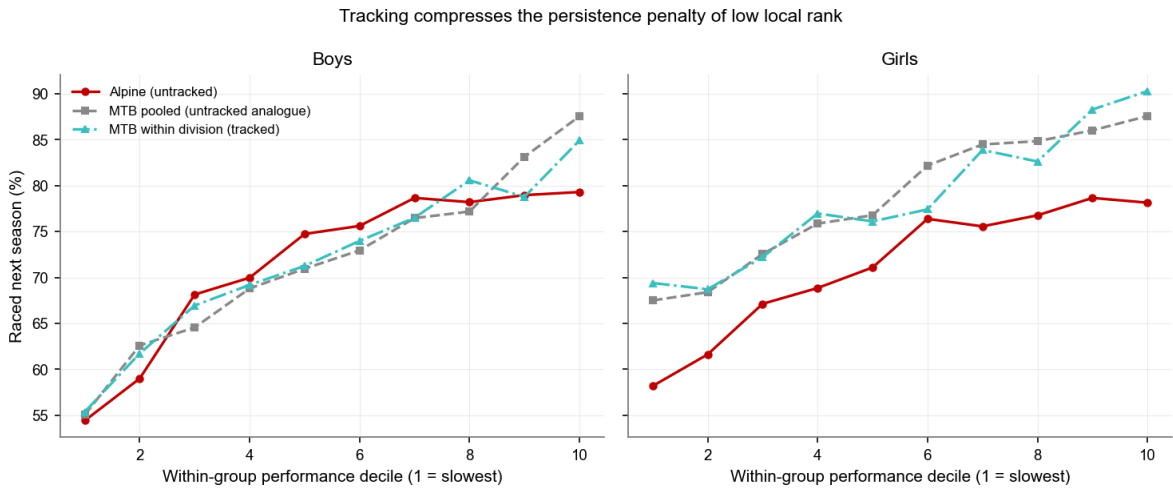


Figure A14: Return rate by performance decile: alpine (untracked) versus MTB

Notes: Harmonized return-by-decile gradients. Alpine: within age-group×gender deciles. MTB: within-division deciles (tracked) and within-grade-gender deciles pooled across divisions (untracked analogue). Raw and ability-adjusted variants; see Table A18.

Table A18: Top-minus-bottom decile return-rate gaps (pp), harmonized

Sport	Series	Boys	Girls
Alpine (untracked)	Raw	24.9	20.0
Alpine (untracked)	Adjusted	14.7	5.6
MTB, pooled grade-gender (untracked analogue)	Raw	32.5	20.1
MTB, pooled grade-gender (untracked analogue)	Adjusted	24.0	12.9
MTB, within division (tracked)	Raw	29.6	20.9
MTB, within division (tracked)	Adjusted	27.8	17.8

Notes: Gap in next-season return rate between the top and bottom performance decile within the indicated grouping. “Adjusted” nets out prior-season ability. The harmonized comparison does not show the tracked gradients to be compressed relative to the untracked sport; see text for caveats (different sports, cost structures, institutions, eras). Descriptive only.